

Pregnancy and Kidney Disease

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CHAPTER OUTLINE

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MASKED HYPERTENSION, 1721	PREGNANCY IN WOMEN RECEIVING KIDNEY REPLACEMENT THERAPY, 1730
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KEY POINTS

- Pregnancy is a “kidney stress test,” which may unmask or worsen preexisting kidney conditions. Vigilance for kidney disease is required, especially in high-risk groups with chronic kidney disease risk factors.
- Parenthood is an important goal for many women with established, known kidney disease, and reproductive counseling is a standard part of best-practice nephrology care.
- Reproductive planning in women at risk of or with kidney disease requires careful planning. Finding the safest window of opportunity for pregnancy requires early implementation of reproductive counseling. This will facilitate shared decision making, individualized risk stratification, optimal timing, and specialized care to optimize the chance of safe and successful pregnancy.
- Preeclampsia is a major risk factor for long-term cardiovascular disease, stroke, kidney failure, and hypertension.
- Women with a history of preeclampsia should be screened and treated for modifiable risk factors for kidney and cardiovascular disease.
- Angiogenic markers have a role in ruling out preeclampsia in women, even those with stable underlying medical disorders.
- Acute kidney injury has myriad causes in pregnancy that vary according to resource and health setting. In many countries, pregnancy-related acute kidney injury is a major cause of long-term kidney failure and maternal mortality.
- Outcomes for women on dialysis or with kidney transplants have improved, but these pregnancies remain complex and high risk, requiring careful planning and management.

Pregnancy is characterized by profound physiologic changes and constitutes a “kidney stress test.” Pregnancy may unmask or accelerate kidney disease or precipitate unique conditions, such as preeclampsia, that have specific kidney impacts short and long term. Significant inequities in health and pregnancy outcomes for women in pregnancy

and women with kidney disease exist in many regions. This chapter discusses the physiologic changes of pregnancy, hypertension in pregnancy including preeclampsia, and best-practice approaches to acute kidney failure, chronic kidney disease of all stages, glomerular disease, and kidney transplantation in pregnancy. Practical frameworks for

preconception counseling, risk assessment, and antenatal and post-partum management for the general nephrologist are provided.

PHYSIOLOGIC CHANGES OF PREGNANCY

HEMODYNAMIC AND VASCULAR CHANGES OF NORMAL PREGNANCY

Normal pregnancy is characterized by profound kidney structural, vascular, hemodynamic, and tubular changes that are essential to maternal-fetal health^{1,2} (summarized in Table 58.1). Systemic vascular resistance (SVR) decreases and arterial compliance increases by 6 weeks' gestation, long before uteroplacental circulation is established.^{3,4} Consequently, mean arterial blood pressure (BP) falls by an average of 10 mm Hg, with nadir at 18 to 24 weeks' gestation. Sympathetic activity is increased, with a 15% to 20% increase in heart rate.⁵ The net effect is increased cardiac output in the early first trimester, which peaks at 50% above prepregnancy levels by the middle of the third trimester (Fig. 58.1).

The renin-aldosterone-angiotensin system (RAAS) is activated in pregnancy,^{2,6} which leads to salt and water retention. Renal interstitial compliance is increased, whereby the renal interstitial pressure remains low despite increased renal interstitial volume. This may contribute to volume retention via an attenuation of the renal pressure natriuretic response.^{7,8} Total body water increases by 6 to 8 L, causing plasma volume and interstitial volume expansion with a fall in osmolality, counteracted by cumulative retention of average of 950 to 1000 mmol of sodium distributed between the maternal extracellular compartments and fetus.⁸ The resultant plasma volume increase leads to mild dilutional anemia. Plasma volume expansion resets the antidiuretic hormone (ADH; also known as "arginine vasopressin") release threshold and increases atrial natriuretic peptide secretion by the late first trimester.^{4,8}

KIDNEY AND HOMEOSTATIC ADAPTATIONS TO NORMAL PREGNANCY

Kidney size and volume increases by 1 to 1.5 cm and up to 30%, respectively.⁹ Asymptomatic, physiologic, mild dilatation of the collecting system occurs in 50% of women, more frequently on the right.¹⁰ These changes may be due to mechanical compression of the ureters by the gravid uterus,¹¹ plus effects of estrogen, progesterone, and prostaglandins on ureteral structure and peristalsis.

Glomerular filtration increases by about 40% to 50% within weeks of conception and is maintained at this level until late pregnancy.¹² In early pregnancy, this is mediated primarily by an increase in renal plasma flow (Fig. 58.2). In the second half of pregnancy, renal plasma flow declines toward prepregnancy levels, yet glomerular filtration rate (GFR) remains high, reflecting relatively increased filtration fraction.² The maintenance of high GFR despite a fall in renal plasma flow is possible due to decreased capillary oncotic pressure and increased K_f (the product of hydraulic permeability and total surface area available for filtration).¹³

Table 58.1 Physiologic Changes in Pregnancy

Physiologic Variable	Change in Pregnancy
Hemodynamic Parameters	
Plasma volume	Increases 30%-50% above baseline due to RAAS activation, progesterone competitively inhibiting aldosterone, reduced set point for ADH release, and sodium retention of up to 1000 mmol total body sodium
Blood pressure	Decreases by approximately 10 mm Hg below prepregnancy level, with nadir in second trimester around 20 weeks Gradually increases toward prepregnancy levels from 35 weeks
Cardiac output	Increases 30%-50%
Heart rate	Increases by 15-20 beats/min
Renal blood flow	Increases to 50%-80% above baseline; afferent and efferent arteriole vasodilation
Glomerular filtration rate	Increased sensitivity to RAAS 150-200 mL/min (increases to 40%-50% above baseline)
Structural changes	Physiologic collecting system dilation and mild hydronephrosis
Serum Chemistry and Hematologic Changes	
Hemoglobin	Decreases by an average of 2 g/L (from 13 to 11 g/L) owing to plasma volume expansion despite increase in red blood cell mass due to increase in erythropoietin
Serum creatinine	Decreases to a nadir of 0.4-0.5 mg/dL in the second trimester and rises back to nonpregnant levels by the third trimester; levels of 0.81-0.87 mg/dL in the second to third trimester represent kidney impairment
Uric acid	Decreases to a nadir of 2.0-3.0 mg/dL by 22-24 weeks due to reduced reabsorption, then increases back to nonpregnant levels toward term
pH	Increases slightly to 7.4-7.45
Partial pressure of carbon dioxide (pCO ₂)	Decreases by approximately 10 mm Hg to an average of 27-32 mm Hg
Calcium	Increased calcitriol stimulates increases in both intestinal calcium reabsorption and increased urinary calcium excretion
Serum sodium	Decreases by 4-5 mEq/L below nonpregnancy levels despite increase in total body sodium
Serum osmolality	Decreases to a new osmotic set point of approximately 270 mOsm/kg with reduced ADH (vasopressin) release
Urinary protein	Physiologic proteinuria up to urine protein-to-creatinine ratio of 30 mg/mmol

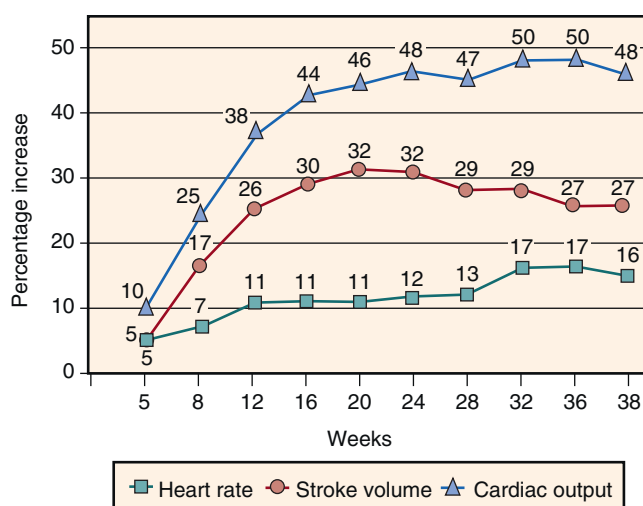


Fig. 58.1 Hemodynamic changes in pregnancy. Shown are the percentage changes from pre-pregnancy values in heart rate, stroke volume, and cardiac output measured throughout pregnancy. (Modified from Robson SC, Hunter S, Boys RJ, et al. Serial study of factors influencing changes in cardiac output during human pregnancy. *Am J Physiol.* 1989;256:H1060-H1065.)

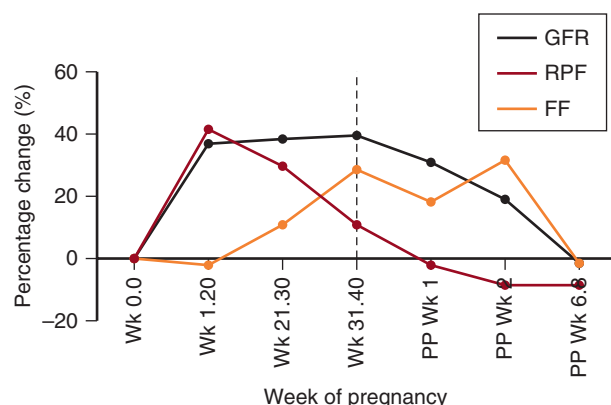


Fig. 58.2 Changes in glomerular filtration (GFR), renal plasma flow (RPF), and filtration fraction (FF) during gestation by week (wk). (From Odutayo A, Hladunewich M. Obstetric nephrology: renal hemodynamic and metabolic physiology in normal pregnancy. *Clin J Am Soc Nephrol.* 2012;7:2073-2080²)

The increase in GFR results in a physiologic decrease in serum creatinine, urea, and uric acid.^{1,2} Normal creatinine clearance in pregnancy rises to 150–200 mL/min (up to 50% increase), and average serum creatinine falls from 0.8 to 0.5 to 0.6 mg/dL by the second trimester.^{12,14} Hence “normal” serum creatinine for a nonpregnant individual reflects kidney impairment in a pregnant woman. Standard estimated GFR equations based on serum creatinine are not validated in pregnancy. The Cockcroft-Gault formula underestimates GFR compared with inulin clearance and overestimates it compared with 24-hour urine creatinine clearance. The Modification of Diet in Renal (Disease and CKD-EPI) formulas underestimate GFR.¹⁵ It is important to note that a small rise in serum creatinine back to prepregnancy levels in the third trimester is normal¹⁴ and does not necessarily reflect kidney impairment. Cystatin C is not routinely used as a reliable marker of kidney function in pregnancy, nor is it widely available.

Physiologic urinary protein excretion rises in normal pregnancy up to 300 mg/day in later pregnancy¹⁶ due to increased GFR, increased permeability of the glomerular basement membrane, and reduced tubular reabsorption of filtered protein.¹³ The higher end of physiologic proteinuria may result in a positive urine dipstick test, particularly on a concentrated urine sample. Women with preexisting proteinuria may have worsening proteinuria in pregnancy, beyond physiologically expected levels.

Due to the large increase in GFR, glomerular tubular balance requires a concomitant increase in tubular solute reabsorption to avoid excessive renal losses. Renal adaptation ensures pregnant women have normal excretion of exogenous solute load and still appropriately conserve sodium when intake is restricted. For example, serum uric acid reaches a nadir of 2.0 to 3.0 mg/dL by 22 to 24 weeks,² returning to nonpregnant levels by term due to increased renal tubular absorption of urate.

The osmotic threshold for stimulation of both ADH release and thirst is decreased, mediated by human chorionic gonadotropin (hCG)¹⁷ and relaxin.¹⁸ This results in mild hyponatremia: The serum sodium typically falls by 4 to 5 mM/L below nonpregnancy levels.

Circulating levels of vasopressinase, which hydrolyzes ADH (arginine vasopressin), are increased during normal pregnancy. Rarely this may lead to a reduction in circulating ADH such that polyuria and polydipsia appear (i.e., transient diabetes insipidus of pregnancy).¹⁹

Pregnancy also causes a progesterone-mediated stimulation of respiratory drive with mild respiratory alkalosis and compensatory renal bicarbonate excretion.¹

MECHANISM OF VASODILATION IN PREGNANCY

Pregnancy induces a widespread decrease in vascular tone, with BP and SVR falling early in pregnancy. Multiple hormones and signaling pathways, including estrogen, progesterone, relaxin, and prostaglandins, contribute to the systemic vasodilatory response. Reduced vascular responsiveness to vasopressors such as angiotensin-2 (AT₂), norepinephrine, and vasopressin in pregnancy all contribute to vascular relaxation in pregnancy.^{6,8,19,20} Relaxin is a corpus luteal hormone that rises early in gestation in response to hCG and may also drive the global and renal vasodilatory response.²¹ Relaxin upregulates intrarenal endothelin and nitric oxide production, leading to generalized renal vasodilation, decreased renal afferent and efferent arteriolar resistance, and increase in renal blood flow and GFR²² (see Table 58.1).

The low-resistance, high-flow circulation of the fetoplacental unit also contributes to the low SVR that is characteristic of the second and third trimesters of pregnancy. During placental development, the high-resistance uterine arteries are transformed into larger-caliber capacitance vessels (Fig. 58.3). This transformation is driven by invasion of the maternal spiral arteries by fetal-derived cytotrophoblasts, which convert from epithelial to endothelial phenotype as they replace the endothelium of the maternal spiral arteries.²³ The mechanisms governing this vascular mimicry or “pseudovasculogenesis” have become clearer in recent years. The dysregulation of angiogenic factors such as vascular endothelial growth

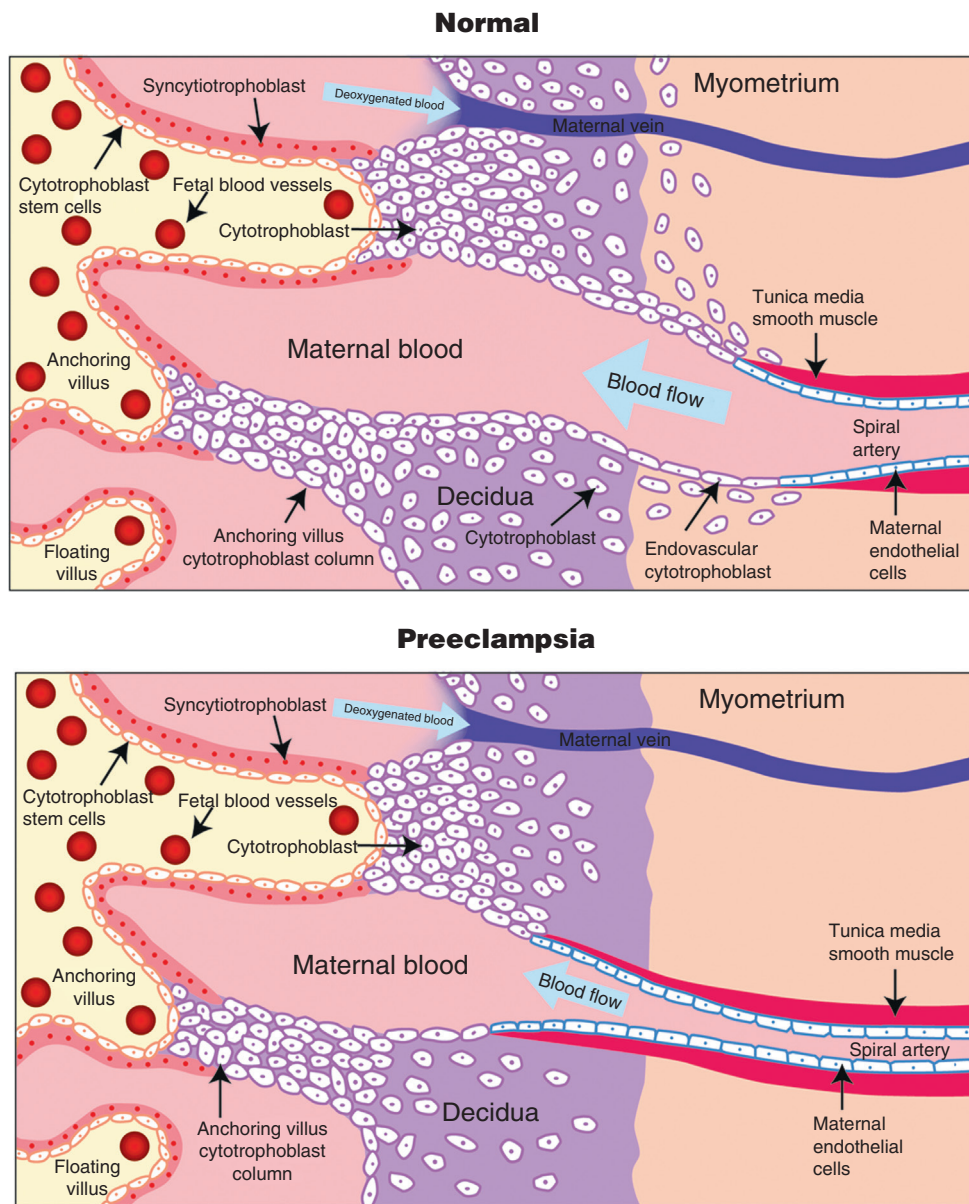


Fig. 58.3 Placentation in normal and preeclamptic pregnancies. In normal placental development (*upper panel*), invasive cytotrophoblasts of fetal origin invade the maternal spiral arteries, transforming them from small-caliber resistance vessels to high-caliber capacitance vessels capable of providing placental perfusion adequate to sustain the growing fetus. During the process of vascular invasion, the cytotrophoblasts differentiate from an epithelial phenotype to an endothelial phenotype, a process referred to as “pseudovasculogenesis” or “vascular mimicry.” In preeclampsia (*lower panel*), cytotrophoblasts fail to adopt an invasive endothelial phenotype. Instead, invasion of the spiral arteries is shallow, and they remain small-caliber resistance vessels. (From Lam C, Kim KH, Karumanchi SA. Circulating angiogenic factors in the pathogenesis and prediction of preeclampsia. *Hypertension* 2005;46:1077–1085.)

factor (VEGF) and angiopoietins has emerged as a key player in understanding placental pathology and its clinical manifestations such as preeclampsia (see “[Pathogenesis of Preeclampsia](#)” later).

HYPERTENSIVE DISORDERS OF PREGNANCY

Hypertension that occurs in pregnancy is best identified as one of the hypertensive disorders of pregnancy. The classification of hypertensive disorders in pregnancy (HDP) is

important as it reflects the pathophysiology of the condition, as well as the risks and potential adverse outcomes for both mother and baby. The HDPs ([Table 58.2](#)) are classified as: Preeclampsia/eclampsia, gestational hypertension (GH), superimposed preeclampsia, and chronic hypertension.

Importantly, other types of hypertension that exist outside of pregnancy may be identified during pregnancy (e.g., white-coat hypertension [WCH] and masked hypertension [MH]). Increasingly, hemolysis, elevated liver enzymes, and low platelet (HELLP) syndrome is being understood to be a severe variant of preeclampsia.

Table 58.2 Definitions of Hypertensive Disorder of Pregnancy Categories and Other Types of Hypertension that May be Identified During Pregnancy

HDP Category	Definition
Preeclampsia ^a	Multisystem disorder. New-onset hypertension (≥ 140 and/or ≥ 90 mm Hg) after 20 weeks' gestation accompanied by ≥ 1 of the following new-onset organ involvement items ^a : - Kidney involvement - Proteinuria (spot urine protein-to-creatinine ratio ≥ 30 mg/mmol [0.3 mg/mg] or >300 mg/day in 24-hr urine collection or albumin-to-creatinine ratio of >8 mg/mmol [71 mg/g]) - Serum creatinine >90 μmol/L OR 1 mg/dL in the absence of other kidney disease - Liver involvement - Raised serum transaminases (in the absence of an alternate diagnosis) with or without right upper quadrant or epigastric pain - Hematologic involvement - Thrombocytopenia $<150,000/\mu$L - Features of hemolysis: decreased haptoglobin with or without fragmented red cells, elevated lactate dehydrogenase - Disseminated intravascular coagulation - Neurologic involvement - Seizure (eclampsia) - Features of cerebral irritability: hyperreflexia with sustained clonus, persistent headache, persistent visual disturbances (photopsia, scotomata, cortical blindness, posterior reversible encephalopathy, retinal vasospasm) - Cerebrovascular accident - Pulmonary edema - Features of placental dysfunction - Features of fetal growth restriction or deceleration in growth trajectory associated with abnormal umbilical artery Dopplers or oligohydramnios
Gestational hypertension ^a	New onset of hypertension after 20 weeks with no evidence of end-organ involvement
Superimposed preeclampsia*	Features of preeclampsia superimposed on either preexisting hypertension or kidney disease (e.g., sudden increase in blood pressure that was well controlled or escalation of antihypertensive medications to control blood pressure, new-onset proteinuria in a woman with chronic hypertension)
Chronic hypertension	Hypertension confirmed either before pregnancy or before 20 completed weeks' gestation. Chronic hypertension may be either: - Primary (or essential) or - Secondary
White coat hypertension	Where hypertension is present in the office or clinic setting but normal blood pressure readings occur at home or with ambulatory blood pressure monitoring
Masked hypertension	Where there is normal blood pressure in the office or clinic setting but hypertension is noted on home readings or with ambulatory blood pressure monitoring

^aThe signs evident of end-organ dysfunction that developed during pregnancy should return to normal within 3 months of birth.

Preeclampsia is a systemic syndrome that is specific to pregnancy, characterized by the new onset of hypertension and evidence of end-organ dysfunction after 20 weeks' gestation. Typically, proteinuria is present, but preeclampsia may manifest as liver, cerebral, hematologic, fetal, or other end-organ dysfunction. Preeclampsia affects approximately 2% to 5% of pregnancies, although there is significant geographic variation in the rate.²⁴ Despite many advances in our understanding of the pathophysiology of preeclampsia, delivery of the placenta and hence the neonate remains the only definitive treatment. The greatest burden of neonatal and maternal mortality and morbidity occurs in less developed nations.²⁵

PATHOGENESIS OF PREECLAMPSIA

ROLE OF THE PLACENTA

The placenta has a central role in preeclampsia. Preeclampsia only occurs in the presence of a placenta, a

fetus is not necessary as is the case of hydatidiform mole.²⁶ Early in pregnancy there is failure of normal pregnancy placental vasculature remodeling, followed by maternal endothelial dysfunction, which manifests as clinically detectable preeclampsia later.

PLACENTAL VASCULAR REMODELING AND ISCHEMIA

Early in normal placental development, extravillous cytotrophoblasts invade the uterine spiral arteries of the decidua and myometrium (see Fig. 58.3). These invasive fetal cells replace the maternal endothelial layer of the uterine vessels, transforming them from small resistance vessels to flaccid, high-caliber capacitance vessels.²⁷ This allows the increase in uterine blood flow needed to sustain the fetus throughout the pregnancy.²⁸ In preeclampsia, this transformation is incomplete.²⁹ The cytotrophoblast invasion of the arteries is limited to the superficial decidua, and the myometrial segments remain narrow and

undilated and hence present high-resistance vessels.³⁰ This has been shown to occur due to a failure of the surface adhesion molecules on the trophoblasts to switch to an invasive phenotype.³¹ Many factors including hypoxia-inducible factor-1 (HIF-1),³² genetic factors (e.g., *STOX-1* gene abnormalities),³³ a deficiency of antiinflammatory imbalance (TH1 dominant vs. TH2 milieu),³⁴ oxidative stress imbalance,³⁵ and reduced circulating PlGF³⁶ in early pregnancy have been proposed.

Normal placentation requires the development of immune tolerance between the fetus and mother. As preeclampsia occurs more often in first pregnancies³⁷ or after a change in partners, it has suggested an etiologic role for abnormal maternal immune response to paternally derived fetal antigens.³⁸

Abnormal uterine artery Doppler ultrasound, consistent with decreased uteroplacental perfusion, is observed before the clinical onset of preeclampsia.^{39,40} The incidence of preeclampsia is increased twofold to fourfold in women residing at high altitudes, implying hypoxia may be a contributing factor.⁴¹ Pregnant subjects with sickle cell disease, who often have pathologic evidence of placental ischemia and infarction, have an increased risk for preeclampsia.⁴² Hypertension and proteinuria can be induced by constriction of uterine blood flow in pregnant primates and other mammals.⁴³ These observations suggest that placental ischemia may be an important trigger for the development of maternal clinical syndrome.

However, placental ischemia may not always be the primary cause of maternal endothelial dysfunction. Several observations call the placental ischemia hypothesis into question in some women with preeclampsia. In a proportion of cases of preeclampsia, there is no evidence of any fetal or placental effects (e.g., fetal growth restriction or abnormal uterine Doppler⁴⁴). The placental histopathology does not reflect the ischemic changes seen in other preeclampsia.⁴⁵ It may be that the placental ischemic damage that more frequently accompanies late-stage preeclampsia may be a secondary event. The primary event may relate to the tolerance of the maternal vasculature to the hemodynamic effects of pregnancy.

Severe preeclampsia is more likely to be associated with pathologic evidence of placental hypoperfusion and ischemia. Placental findings include acute atherosclerosis, a lesion of diffuse vascular obstruction that includes fibrin deposition, intimal thickening, necrosis, atherosclerosis, and endothelial damage. Infarcts, likely due to occlusion of maternal spiral arteries, are also common.⁴⁶ Although these findings are not universal, they appear to be correlated with severity of clinical disease.⁴⁷

MATERNAL ENDOTHELIAL DYSFUNCTION

The systemic multiorgan clinical manifestations of preeclampsia reflect widespread endothelial dysfunction, resulting in vasoconstriction and end-organ ischemia.^{48,49} Dozens of serum markers of endothelial activation may be deranged.^{49,50} There is evidence for oxidative stress and platelet activation,⁵¹ prostaglandin dysregulation,⁵² TH2 inflammatory cytokine upregulation, and neutrophil infiltration.^{53,54} Several of these aberrations occur well

before the onset of symptoms, supporting the central role of endothelial dysfunction in the pathogenesis of preeclampsia.

HEMODYNAMIC CHANGES IN PREECLAMPSIA

In preeclampsia, SVR is generally high and cardiac output is low,⁵⁵ reflecting widespread vasoconstriction and impaired endothelium-dependent vasorelaxation, which has been noted prospectively before onset of hypertension and proteinuria⁵⁶ and persists for years after the preeclampsia episode.⁵⁷ Subtle increases in BP and pulse pressure are present before the onset of overt hypertension preeclampsia, suggesting that arterial compliance is decreased early in the course of the disease.^{58,59}

ANGIOGENIC IMBALANCE IN PREECLAMPSIA

Angiogenic factors are likely to be important in the regulation of placental vasculogenesis. Epidemiologic studies and experimental studies in animals suggest that excess placental production of soluble VEGFR-1, referred to as “soluble fms-like tyrosine kinase-1” (sFLT1, or sVEGFR-1), plays a causal role in mediating the signs and symptoms of preeclampsia.⁶⁰ sFLT1, a truncated splice variant of the VEGFR Flt1, antagonizes VEGF and PlGF by binding them in the circulation and preventing interaction with their endogenous receptors in the vasculature (Fig. 58.4). sFLT1 inhibits VEGF- and PlGF-mediated angiogenesis and is upregulated in the placenta of women with preeclampsia, resulting in elevated circulating levels. The increase in maternal circulating sFLT1 precedes the onset of clinical disease (Figs. 58.5 and 58.6).^{36,61} Increased circulating sFLT1 is accompanied by decreased circulating free PlGF in serum. In vitro effects of sFLT1 include vasoconstriction and endothelial dysfunction. Exogenous or endogenously produced sFLT1 administered to pregnant rats produces a syndrome

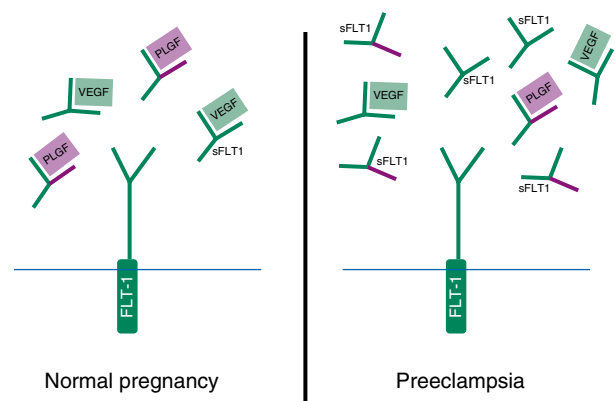


Fig. 58.4 Proposed mechanism of soluble fms-like tyrosine kinase-1 (sFLT1)-induced endothelial dysfunction. sFLT1 protein, derived from alternative splicing of FLT1, lacks the transmembrane and cytoplasmic domains but still has the intact vascular endothelial growth factor (VEGF) and placental growth factor (PlGF) binding extracellular domain. During normal pregnancy, VEGF and PlGF signal through the VEGF receptors (FLT1) and maintain endothelial health. In preeclampsia, excess sFLT1 binds to circulating VEGF and PlGF, thus impairing normal signaling of both VEGF and PlGF through their cell surface receptors. Thus excess sFLT1 leads to maternal endothelial dysfunction.

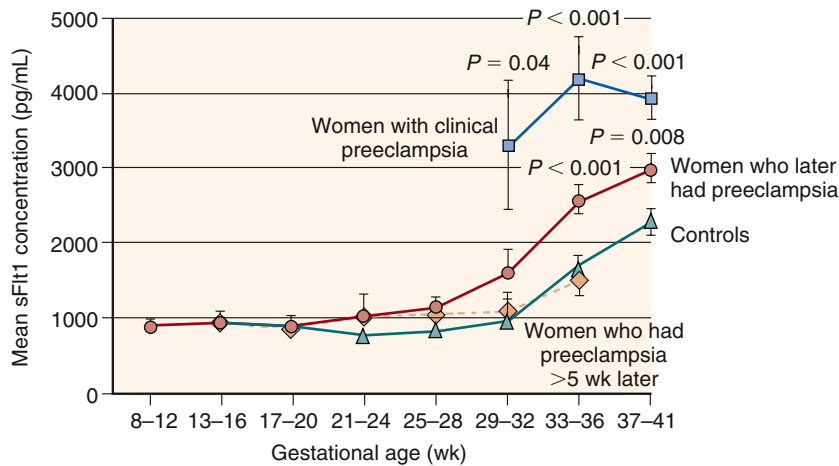


Fig. 58.5 Concentrations of soluble fms-like tyrosine kinase-1 (sFLT1) in preeclampsia and normal pregnancy. Shown are the mean serum sFLT1 concentrations ($\pm$ standard error of mean) before and after onset of clinical preeclampsia according to the gestational age of the fetus. The *P* values given are for comparisons, after logarithmic transformation, with specimens from controls obtained during the same gestational-age interval. All specimens were obtained before labor and delivery. (From Levine RJ, Maynard SE, Qian C, et al. Circulating angiogenic factors and the risk of preeclampsia. *N Engl J Med*. 2004;350:672–683.)

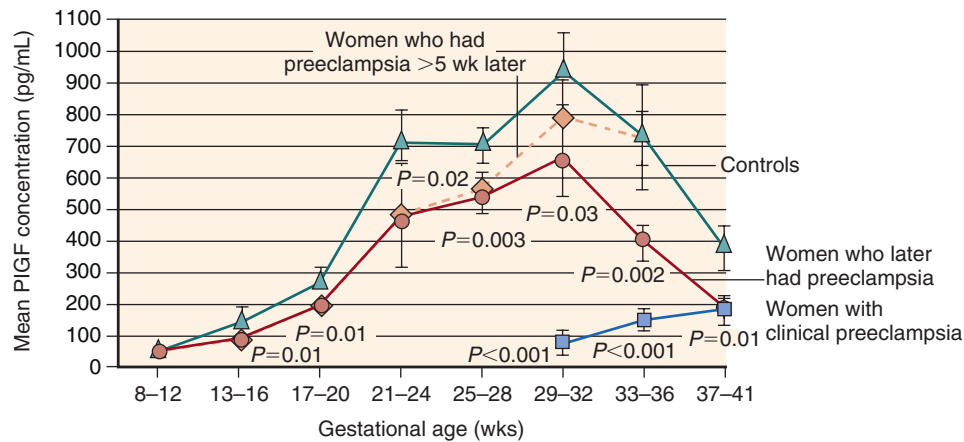


Fig. 58.6 Concentrations of placental growth factor (PlGF) in preeclampsia and normal pregnancy. Shown are the mean serum PlGF concentrations ($\pm$ standard error of the mean) before and after onset of clinical preeclampsia according to the gestational age of the fetus. The *P* values given are for comparisons, after logarithmic transformation, with specimens from controls obtained during the same gestational-age interval. All specimens were obtained before labor and delivery. (From Levine RJ, Maynard SE, Qian C, et al. Circulating angiogenic factors and the risk of preeclampsia. *N Engl J Med*. 2004;350:672–683.)

resembling preeclampsia, including hypertension, proteinuria, and glomerular endotheliosis.^{60,62} The preeclampsia-like syndrome induced by sFLT1 or placental ischemia in animals can be rescued by exogenous VEGF or PlGF administration.^{63,64}

Derangements in other angiogenic molecules have also been observed including endostatin,⁶⁵ VEGF165b, and circulating levels of soluble endoglin (sEng).⁶⁶ Maternal serum levels of sEng rise before preeclampsia onset,⁶⁷ in a pattern similar to sFLT1.⁶⁸ Higher sFLT1 levels have been noted in first versus second pregnancies⁶⁹ in twin versus singleton pregnancies⁷⁰ in women with prior preeclampsia⁷¹ and in women carrying fetuses affected by trisomy 13.⁷² In the case of twin pregnancies, the increased sFLT1 production appears to be due to increased placental mass, rather than placental ischemia. Conversely, lower levels of sFLT1 in pregnant smokers have been generally noted^{73,74} and may explain the protective effect of smoking observed

on preeclampsia risk mediated through the effects of smoking on other molecules.⁷⁵

VEGF ligands and receptors are highly expressed by placental tissue in the first trimester. sFLT1 decreases cytotrophoblast invasiveness in vitro.⁷⁶ Animal models of preeclampsia based on induction of uteroplacental ischemia are characterized by increased endogenous sFLT1⁴³ and sEng.⁷⁷ This supports the two-stage theory of the disease where the placental dysfunction, manifest in part as placental ischemia or under perfusion, results in increased circulating sFLT1 seen in the women who will develop and have developed preeclampsia.

In addition to angiogenic alterations, women who develop preeclampsia also have evidence of insulin resistance.⁷⁸ Moreover, women with pregestational or gestational diabetes mellitus have an increased risk for developing preeclampsia.⁷⁹ In accordance with this finding, in vitro models suggest that insulin signaling and angiogenesis are

intimately related at a molecular level,⁸⁰ and epidemiologic data find that evidence of altered angiogenesis and excess insulin resistance may be additive insults that lead to preeclampsia. Furthermore, altered levels of biomarkers linked with angiogenesis and insulin resistance persist in the postpartum state,⁸¹ possibly explaining the long-term cardiovascular risk in these women.

PATHOLOGIC FINDINGS IN PREECLAMPSIA

KIDNEY CHANGES IN PREECLAMPSIA

The pathologic swelling of glomerular endothelial cells in preeclampsia was first described in 1924. Thirty-five years later, Spargo and colleagues⁸² coined the term “glomerular endotheliosis” and characterized ultrastructural changes, including generalized swelling and vacuolization of the endothelial cells (Fig. 58.7).⁸³ There are deposits of fibrinogen and fibrin within and under the endothelial cells and loss of glomerular endothelial fenestrae. Occasionally, paramesangial deposition of fibrinoid material and mesangial expansion can be identified.^{83,84}

The primary injury is specific to endothelial cells; however, as a consequence, the podocytes are also affected. There is a downregulation of nephrin and synaptopodin in the podocytes.⁸⁵ Serum from preeclamptic women reduces nephrin expression by cultured podocytes.⁸⁶ Using a mouse model of preeclampsia induced by administration of anti-VEGF antibodies, it was noted that nephrin expression is reduced in podocytes.⁸⁷ The effect of sFLT1 on podocyte nephrin expression appears to be mediated via increased endothelin-1 release by endothelial cells.⁸⁶ Thus the reduction in podocyte-produced VEGF may be exacerbated by the increase in circulating sFLT1. This affects the fenestrations in the endothelial cells, which then produces endothelin-1, which induces nephrin shedding and hence the shedding of podocytes into the urine⁸⁸ (Fig. 58.8).

Both renal blood flow and GFR are low in preeclampsia compared with normal pregnancy. Renal blood flow falls as a result of high renal vascular resistance, primarily due to increased afferent arteriolar resistance and reduced cardiac output.⁸⁹ GFR falls as a result of both the fall in renal blood flow and a decrease in the ultrafiltration coefficient (Kf), attributed to endotheliosis in the glomerular capillary.⁹⁰ In extremes of preeclampsia, AKI can occur due to the reduction in glomerular perfusion and eventually irreversible loss of kidney function where cortical necrosis occurs.⁹¹

CEREBRAL CHANGES IN PREECLAMPSIA

Cerebral edema and intracerebral parenchymal hemorrhage are common autopsy findings in women who died from eclampsia.⁹² The presence of cerebral edema in eclampsia correlates with markers of endothelial damage but not the severity of hypertension,⁹³ suggesting that the edema is secondary to endothelial dysfunction rather than a direct result of BP elevation. Findings on head computed tomography (CT) and magnetic resonance imaging (MRI) are similar to those seen in hypertensive encephalopathy, with vasogenic cerebral edema and infarctions in the subcortical white matter and adjacent gray matter, predominantly in the parieto-occipital lobes.⁹⁴ The syndrome (reversible posterior leukoencephalopathy)

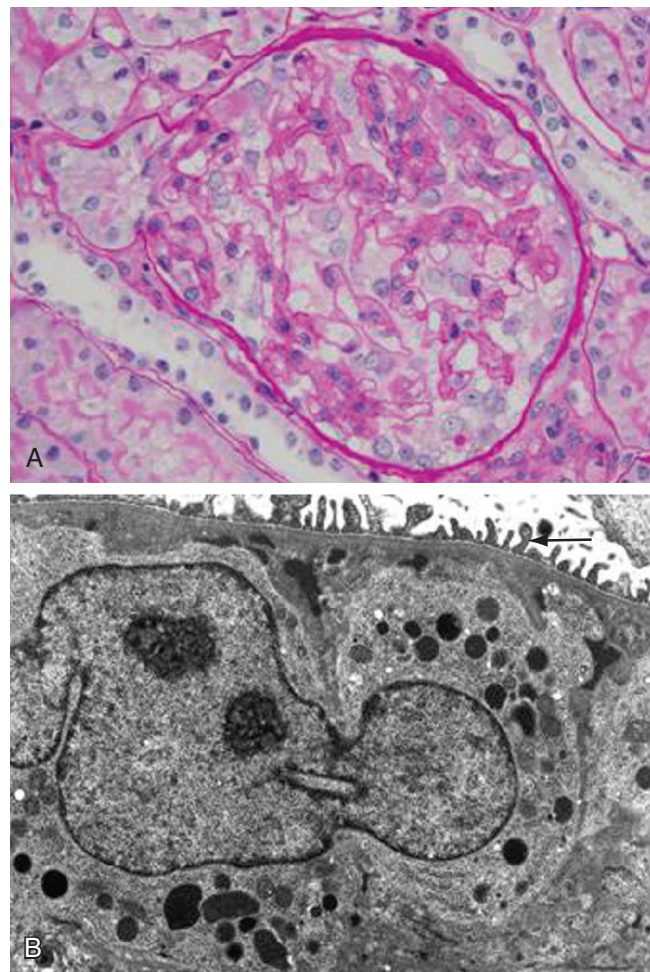


Fig. 58.7 Glomerular endotheliosis. (A) Human preeclamptic glomerulus on light microscopy (periodic acid-Schiff stain). Renal biopsy findings from a 29-year-old woman with twin gestation and severe preeclampsia are shown. The patient's blood pressure was 170/112 mm Hg, and random urine protein-to-creatinine ratio was 9.8. Note the “bloodless” appearance of the glomeruli and absence of the capillary lumen. (Original magnification $\times 40$.) (B) Electron microscopy of biopsy specimen of the glomerulus from the same patient. Note occlusion of capillary lumen cytoplasm and expansion of the subendothelial space with some electron-dense material. Podocyte cytoplasm shows protein resorption droplets and relatively intact foot processes. (Original magnification $\times 1500$.) (Courtesy I. E. Stillman.)

includes these characteristic MRI changes, together with headaches, seizures, altered mental status, and hypertension. It has been described in patients with acute hypertensive encephalopathy in the setting of kidney disease, eclampsia, or immunosuppression.

LIVER CHANGES IN PREECLAMPSIA

Similarly, the primary pathogenic issue in the liver is endothelial dysfunction, with microvesicular, ischemic lesions, simultaneous periportal hemorrhage,⁹⁵ and necrosis in the midzones of lobules, which may become confluent.⁹⁶ Small thrombi may be seen in hepatic arterioles and ultimately a subcapsular or intrahepatic hematoma may develop and in the extreme lead to liver rupture.⁹⁷ Causative factors are the imbalance of angiogenic factors, mitochondrial dysfunction, and oxidative stress in hepatic

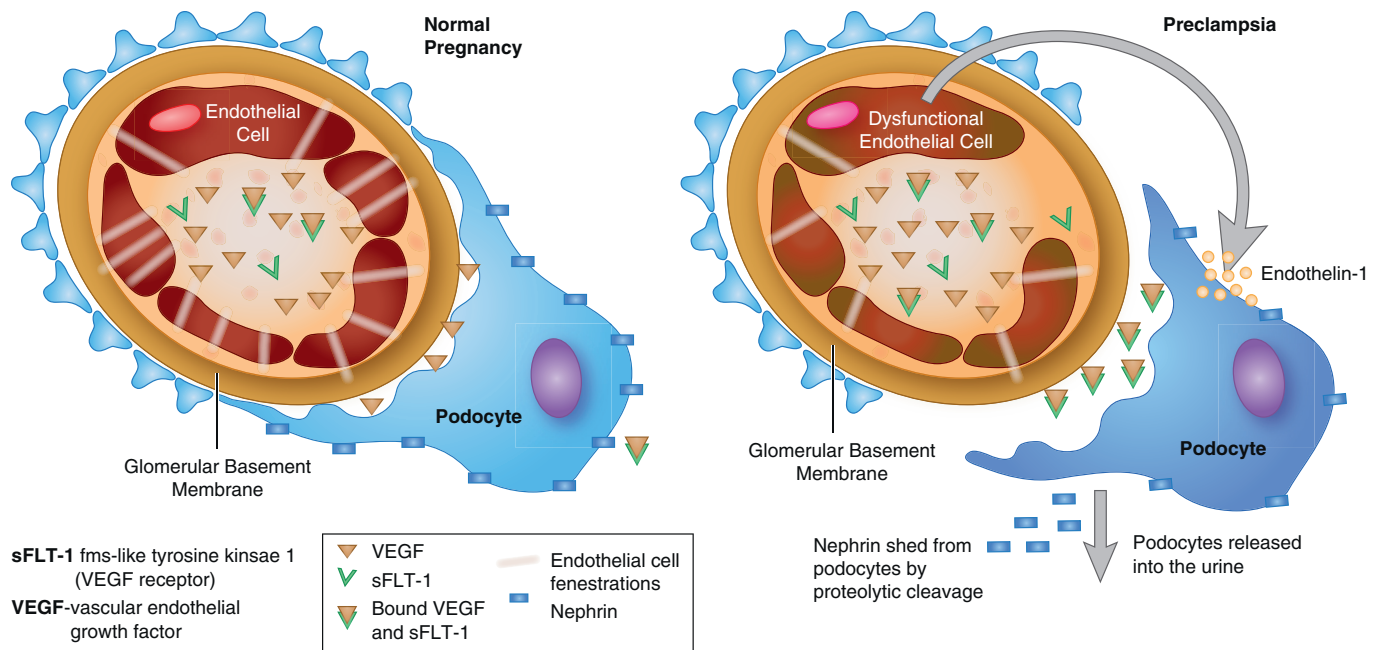


Fig. 58.8 Relationship between podocyte dysfunction and glomerular endothelial cell distribution in preeclampsia. Schema summarizing changes in the glomerulus barrier in normal and preeclamptic pregnancy. (A) In a normal pregnancy, there is an excess of vascular endothelial growth factor VEGF (V shape), compared with the circulating fms-like tyrosine kinase-1 (sFLT-1; black triangle) in women who will develop preeclampsia. (B) Placentally derived factors cause endothelial dysfunction, which does not have a direct effect on the podocytes. The circulating sFLT-1 does bind to and inhibit the effect of the podocyte-produced VEGF thought to have a major role in maintaining a healthy glomerular barrier. Dysfunctional endothelial cells produce endothelin-1 (ET, red circles), which in turn affects the podocytes and induces nephrin shedding (red rectangle). The placentally derived factors cause a reduction in endothelial cell fenestrations (straight lines). (Adapted from Hennessy A, Makris A. Preeclamptic nephropathy. *Nephrology* (Carlton). 2011;16[2]:134–143.⁸⁸

organelles.⁹⁸ The changes can make it hard to identify women with preeclampsia with hepatic involvement from women who have acute fatty liver of pregnancy⁹⁹ (see “[Acute Fatty Liver of Pregnancy](#)” later).

HEMATOLOGIC CHANGES IN PREECLAMPSIA

Women with preeclampsia develop significant thrombocytopenia, which can progress to a severe reduction in circulating platelet numbers. This is thought to occur because of platelet consumption as a result of platelet activation due to endothelial dysfunction.¹⁰⁰ The ongoing microangiopathy results in hemolysis, and eventually disseminated coagulation can also occur.

PREECLAMPSIA EPIDEMIOLOGY AND RISK FACTORS

The incidence of preeclampsia varies among populations. Most cases of preeclampsia occur in healthy nulliparous women in whom the incidence of preeclampsia has been reported as high as 7.5%.¹⁰¹ While classically a disorder of first pregnancies, multiparous women who are pregnant with a new partner appear to have an elevated preeclampsia risk similar to that of nulliparous women.¹⁰² This effect may be due to increased interpregnancy interval rather than the change in partner.^{103,104} Similarly, women who are pregnant at an advanced maternal age (>40 years) have a higher risk of preeclampsia.¹⁰⁵

Although most cases of preeclampsia occur in the absence of a family history, the presence of preeclampsia in a first-degree relative increases a woman's risk of

severe preeclampsia twofold to fourfold,¹⁰⁶ suggesting a genetic contribution to the disease. However, to-date results from genome-wide scans seeking a specific linkage to preeclampsia have not identified specific genetic mutations.

Several medical conditions are associated with increased preeclampsia risk including chronic hypertension, diabetes mellitus (types 1 and 2), obesity, and antiphospholipid antibody syndrome (Table 58.3).¹⁰⁵ Women with preeclampsia in a prior pregnancy have a high risk of preeclampsia in subsequent pregnancies. Women with early preterm preeclampsia have the highest rate of recurrence. Pregnancies that occur in adolescent girls also have a high risk of preeclampsia.¹⁰⁷ Adults who were born with low birth weights are at increased risk of pregnancy complications including preeclampsia and GH (for further discussion see Chapter 20).¹⁰⁸

Conditions associated with increased placental mass, such as multifetal gestations and hydatidiform mole, are also associated with increased preeclampsia risk. Trisomy 13 is associated with a high risk of preeclampsia.⁷² Reduced paternal sperm exposure and shorter sexual cohabitation have a higher risk of preeclampsia.¹⁰⁹ Sperm exposure orally similarly reduces the risk of preeclampsia.¹¹⁰ Pregnancies requiring assisted reproductive techniques (ARTs), such as in vitro fertilization/intracytoplasmic sperm injection, compared with spontaneous pregnancies have a higher rate of preeclampsia (odds ratio [OR] 1.59, 95% confidence interval [CI] 1.46–1.73). Interestingly, pregnancies with oocyte donation had the highest rate of preeclampsia (OR 4.96 [95% CI 3.52–7.0]). More recently frozen embryo

Table 58.3 Risk Factors for Preeclampsia

Risk Factor	Pooled Relative Risk (95% CI)
Antiphospholipid antibody syndrome	2.8 (1.8–4.3) ^a
Chronic kidney disease	1.8 (1.5–2.1) ^a
Prior preeclampsia	8.4 (7.1–9.9) ^a
Nulliparity	2.1 (1.9–2.4) ^a
Chronic hypertension	5.1 (4.0–6.5) ^a
Pregestational diabetes mellitus	3.7 (3.1–4.3) ^a
Multiple gestations	2.9 (2.6–3.1) ^a
Strong family history of cardiovascular disease (heart disease/stroke in two or more first-degree relatives)	3.2 (1.4–7.7) ^b
Systemic lupus erythematosus	2.5 (1.0–6.3) ^a
Obesity (pregnancy BMI >30 kg/m ²)	2.8 (2.6–3.1) ^a
Family history of preeclampsia	2.4 (1.8–3.6) ^c
Advanced maternal age (>40 years)	1.5 (1.2–2.0) ^a
Excessive gestational weight gain (>35 lb)	1.9 (1.7–2.0) ^d
Assisted reproductive technology	1.8 (1.6–2.1) ^a
Previous episode of acute kidney injury	5.9 (3.6–9.7) ^e

^aFrom Bartsch E, Medcalf KE, Park AL, et al. High Risk of Pre-eclampsia identification G. Clinical risk factors for pre-eclampsia determined in early pregnancy: systematic review and meta-analysis of large cohort studies. *BMJ*. 2016;353:i1753.

^bFrom Ness RB, Markovic N, Bass D, et al. Family history of hypertension, heart disease, and stroke among women who develop hypertension in pregnancy. *Obstet Gynecol*. 2003;102:1366–1371.

^cFrom Carr DB, Epplein M, Johnson CO, et al. A sister's risk: family history as a predictor of preeclampsia. *Am J Obstet Gynecol*. 2005;193: 965–972.

^dFrom Macdonald-Wallis C, Tilling K, Fraser A, et al. Gestational weight gain as a risk factor for hypertensive disorders of pregnancy. *Am J Obstet Gynecol*. 2013;209:327 e1–e17.

^eFrom Tangren JS, Powe CE, Ankers E, et al. Pregnancy outcomes after clinical recovery from AKI. *J Am Soc Nephrol*. 2017;28:1566–1574.

transfers have higher odds of preeclampsia compared with fresh embryo transfers.¹¹¹

Several putative risk factors remain controversial. Congenital or acquired thrombophilia is associated with preeclampsia in some but not all studies.¹⁰⁵ Racial differences in the incidence and severity of preeclampsia have been difficult to assess due to confounding by socioeconomic and cultural factors. Although population-based studies have reported a higher rate of preeclampsia among black women¹¹² and Pacific Islanders,¹¹³ these findings have not been confirmed in studies confined to healthy, nulliparous women. This suggests that the increased preeclampsia incidence noted in some studies may be attributable to the

higher rate of chronic hypertension in these populations, as chronic hypertension is itself a strong risk factor for preeclampsia.¹¹⁴

The longer duration of hypertension before pregnancy and the lack of treatment for prepregnancy chronic hypertension are further risk factors for preeclampsia in women with chronic hypertension. However, the effect of prepregnancy BP on pregnancy is a continuum, with evidence that prehypertension (systolic and/or diastolic hypertension) increases the risk of preeclampsia.¹¹⁵

PREECLAMPSIA DIAGNOSIS AND CLINICAL FEATURES

In recent years there has been a greater concordance between the definitions of preeclampsia across international groups. Generally, a diagnosis of preeclampsia may be made with the new inset of hypertension and evidence of end organ dysfunction (see Table 58.2). The requirement of the presence of proteinuria is no longer necessary. Historically, edema was considered part of the clinical presentation or preeclampsia. However, edema is present in 60% to 80% of normal pregnancies¹¹⁶ and, as such, it is no longer required due to its lack of specificity and sensitivity for the diagnosis of preeclampsia. Similarly, an elevated serum uric acid was considered important for the diagnosis of preeclampsia. The serum uric acid is positively correlated with the severity of preeclampsia.¹¹⁷ However, it is of limited clinical utility in distinguishing preeclampsia from other HDPs or as a predictor of adverse outcomes. Similarly, it has no role in the decision to deliver a woman with preeclampsia.

HYPERTENSION

For the diagnosis of preeclampsia, hypertension is defined as systolic blood pressure (SBP) of ≥ 140 mm Hg and/or diastolic blood pressure (DBP) after 20 weeks' gestation in a woman with previously normal BP. The BP readings should be confirmed on at least 2 occasions, generally at least 4 hours apart. The sphygmomanometer should be a machine that is validated for use in pregnancy. In women with chronic hypertension, a sudden increase in BP or escalation of antihypertensive medications required to control BP may indicate that preeclampsia may be superimposed on the preexisting hypertension.

PROTEINURIA

Proteinuria is a hallmark of preeclampsia. For the diagnosis of preeclampsia, proteinuria is defined as >300 mg protein in a 24-hour urine collection or a spot urine protein-to-creatinine (P:C) ratio >0.3 mg/mg (>30 mg/mmol) or urine dipstick of $\geq 2+$ (if quantitative methods are unavailable). However, preeclampsia can be diagnosed even in the absence of proteinuria, if other features are present. Routine obstetric care includes dipstick protein testing of a random voided urine sample at each prenatal visit; however, this has been shown to have a high rate of false-positive and false-negative results when compared with 24-hour urine protein measurement or spot P:C ratio.¹¹⁸ Although the 24-hour urine collection for proteinuria is considered the gold standard in assessing proteinuria, issues related to collection and delay in obtaining results mean that the spot P:C ratio has become the preferred method

for quantification of proteinuria.¹¹⁹ An alternate assessment of proteinuria in pregnancy, albuminuria has been less better studied in preeclampsia than other kidney disorders. A small number of studies have investigated the equivalent albuminuria to the already existing cutoff of pregnancy-related proteinuria (>300 mg/24 hours, spot urine P:C ratio of >0.3 mg/mg [30 mg/mmol]). A spot urinary cutoff of an albumin-to-creatinine (A:C) ratio of ≥ 0.08 mg/mg (8 mg/mmol) would have an acceptable sensitivity and sensitivity compared with the P:C ratio.¹²⁰ Generally, laboratory testing for albuminuria is more costly and less widely available than assays for proteinuria.

The degree of proteinuria in preeclampsia can range widely, but the degree of proteinuria is a poor predictor of adverse maternal and fetal outcomes¹²¹ and alone is not an indication for urgent delivery. Among women with underlying proteinuria, other signs of preeclampsia such as abnormal angiogenic biomarkers, elevated transaminases, thrombocytopenia, or cerebral signs/symptoms are more useful to diagnose superimposed preeclampsia.

ANGIOGENIC MARKERS FOR PREECLAMPSIA DIAGNOSIS

Significant elevations in maternal sFLT1 and sEng are observed from midgestation onward^{61,67,122,123} and appear to rise 5 to 8 weeks before preeclampsia onset (see Figs. 58.5 and 58.6).³⁶ Maternal sFLT1 levels are particularly elevated in severe preeclampsia, early-onset preeclampsia, and preeclampsia complicated by a small-for-gestation infant.¹²⁴ Serum levels of PIGF are lower in women who go on to develop preeclampsia from the first¹²⁵ or early second trimester.¹²⁶ Since PIGF passes into the urine, low urinary PIGF has been identified as a potential marker for preeclampsia. Urinary levels of PIGF are significantly lower in women who develop preeclampsia from the late second trimester¹²⁷ but have not been useful for preeclampsia screening.

Some studies have suggested that circulating angiogenic factors in plasma or serum can be used to differentiate preeclampsia from other diseases that mimic preeclampsia such as chronic hypertension, GH, lupus nephritis, and CKD.¹²⁸⁻¹³¹ Zeisler and colleagues¹³² demonstrated in a prospective multicenter clinical trial that serum sFlt/PIGF can be used to rule out preeclampsia among patients with suspected disease with negative predictive value >99%. Several groups have also demonstrated a role for angiogenic biomarkers in the prediction of preeclampsia-related adverse outcomes among women evaluated for suspected preeclampsia.^{133,134} Measurements of circulating angiogenic factors (sFLT1, PIGF, and sEng) robustly predicted adverse maternal and perinatal outcomes in women presenting with signs or symptoms of preeclampsia, and these biomarkers outperformed the standard battery of clinical diagnostic measures including BP, proteinuria, uric acid, and other laboratory assays.¹³⁵ Although sFLT1 and/or PIGF levels at presentation were associated with the remaining duration of pregnancy, there are no data to support its use for deciding on timing of delivery. There is increasing understanding of the physiology of these markers (e.g., the understanding that circulating sflt-1 and PIGF are affected by other agents such as heparin¹³⁶ and magnesium¹³⁷). By providing more accurate identification of women at high risk for adverse

Table 58.4 Comparison of Clinical and Laboratory Characteristics of the Typical Clinical Syndromes for PE/HELLP, HUS/TTP, and AFLP

	HUS/TTP	PE/HELLP	AFLP
Clinical Characteristic:			
Hemolytic anemia	+++	++	±
Thrombocytopenia	+++	++	– or +
Coagulopathy	–	– or +	+
Neurologic symptoms	++	– or +	– or +
Acute kidney injury	+++	+	++
severity			
Hypertension	– or +	+++	– or +
Proteinuria	– or +	++	– or +
Elevated AST	– or +	++	+++
Elevated bilirubin	++	+	+++
Anemia	++	+	– or +
Elevated ammonia	–	–	+++

AFLP, Acute fatty liver of pregnancy; AST, aspartate aminotransferase; HELLP, hemolytic anemia, elevated liver enzymes, and low platelets; HUS/TTP, hemolytic uremic syndrome/thrombotic thrombocytopenic purpura; PE, Preeclampsia; –, absent; +, mild; ++, moderate; +++, severe or a prominent feature.

pregnancy outcomes, use of angiogenic biomarkers has been shown to reduce costs and unnecessary resource use in women with possible preeclampsia.^{138,139}

HELLP SYNDROME

“HELLP” is an acronym for the syndrome of hemolytic anemia, elevated liver enzymes, and low platelets. There remains considerable variability regarding diagnostic criteria for the HELLP syndrome in the medical literature. It is generally considered to be part of the preeclampsia (severe form) spectrum of disorders rather than its own specific entity. As its name suggests, a diagnosis of HELLP syndrome is suggested by evidence of hemolytic anemia, thrombocytopenia (platelet count <100,000/ μ L), and impaired liver function (liver enzyme levels more than twice the upper limit of normal).¹⁴⁰ This syndrome is sometimes challenging to distinguish from thrombotic thrombocytopenic purpura (TTP), hemolytic uremic syndrome (HUS), and acute fatty liver of pregnancy (AFLP), which can present similarly (Table 58.4). The HELLP syndrome is associated with particularly high rates of adverse maternal and neonatal outcomes including eclampsia (affecting 6% of cases), placental abruption (10%), acute kidney failure (5%), disseminated intravascular coagulation (8%), pulmonary edema (10%),¹⁴¹ and (rarely) hepatic hemorrhage and rupture.¹⁴² Angiogenic markers are abnormal in women with HELLP and thus will not help differentiate it from preeclampsia or AFLP.¹⁴³

POSTPARTUM RECOVERY AFTER PREECLAMPSIA

Generally, preeclampsia begins to remit soon after delivery of the fetus and placenta and complete recovery is the rule. However, normalization of BP and proteinuria often takes

Table 58.5 Oral Antihypertensives That Can be Used During Pregnancy

Antihypertensives	Class of Agent	Dose (Start From Low Dose and Titrate as Required)	Caution	Side Effects
Oral methyldopa	α Blocker	250-750 mg 3-4 times a day	Avoid in women with a history of depression, anxiety, or postpartum depression	Dry mouth, sedation, abnormal liver function tests, hemolytic anemia
Oral clonidine	α Blocker	75-300 micrograms 3-4 times a day	Risk of rebound hypertension with sudden withdrawal	Dry mouth, sedation
Oral labetalol	α And β blocker	100-400 mg 3-4 times a day	Avoid in women with a history of asthma or chronic airway limitation	Theoretic benefit to uteroplacental perfusion compared with other β blockers
Oral oxprenolol	β Blocker	40-120 mg 3 times a day	Avoid in women with a history of asthma or chronic airway limitation	Bradycardia, fatigue
Oral nifedipine SR	Calcium channel blocker	20-60 mg (slow release) twice a day	Avoid in women with aortic stenosis, may cause peripheral edema	Peripheral edema, headaches, palpitations
Oral nifedipine IR	Calcium channel blocker	10-30 mg (immediate release) 3 times a day	Avoid in women with aortic stenosis, may cause peripheral edema	Peripheral edema, headaches, palpitations
Oral hydralazine	Vasodilator	12.5-50 mg 3-4 times a day	May cause tachycardia if given as first line (without concurrent α , β , or calcium blockade)	Tachycardia, headache, lupus-like syndrome, sweating, shortness of breath

IR, Immediate release; SR, sustained release.

days to weeks and generally any clinical findings that persist beyond 3 months postpartum should prompt investigation. Management of hypertension postpartum is similar to the management during pregnancy (Table 58.5). However, as the fetus is delivered, there are some additional treatment options available.

Several studies have shown that loop diuretics (furosemide [20–40 mg] or other pregnancy-safe diuretics) in the short term (up to 5 days postpartum) can be used safely in the management of hypertension.¹⁴⁴⁻¹⁴⁷ These studies have shown a reduction in the development of persistent hypertension as measured on discharge and an increase in the rate of fall in the BP during hospitalization. There was no increase in the rate of acute kidney injury or breastfeeding-related issues. It may be beneficial to start these agents for treating hypertension in the postpartum compared with other agents.^{148,149}

Similar to antenatally, postnatally there is a lack of data on the superiority of any class of antihypertensive agents compared with another. However, it is important to note that angiotensin-converting enzyme inhibitors (ACEIs) can

be used in the postpartum period with some consideration of the excretion of these agents in the breast milk. Enalapril, captopril, and quinapril or their metabolites have been shown to have minimal or no excretion in breastmilk and, as such, can be used for the treatment of hypertension in the postpartum.^{148,150}

Lastly, nonsteroidal antiinflammatory drugs (NSAIDs) are used routinely after birth as they have demonstrated excellent analgesic efficacy.¹⁵¹ However, in the context of women with already elevated BP (regardless of the type of HDP), it is generally suggested that the routine use of NSAIDs be avoided.

LONG-TERM CARDIOVASCULAR AND KIDNEY OUTCOMES AFTER PREECLAMPSIA

It is now well described that although the clinical signs of preeclampsia and GH may remit soon after delivery, women who developed these disorders have an increased risk of longer-term cardiovascular, cerebrovascular, and kidney disease.¹⁵² This is above and beyond the increased

risk of pregnancy complications in any future pregnancy (e.g., preeclampsia). Women who have experienced severe preeclampsia, recurrent preeclampsia, preeclampsia with preterm birth, and preeclampsia with intrauterine growth restriction (IUGR) are most strongly associated with the remote consequences linked to adverse pregnancy outcomes.^{153,154}

As early as 1 year after the affected pregnancy, women with preeclampsia have an increased risk of hypertension, obesity, hypercholesterolemia, microalbuminuria, and new-onset diabetes mellitus, when compared with control women who have not had preeclampsia.¹⁵⁵⁻¹⁵⁷ Relative risk (RR) of ischemic heart disease, stroke, cardiomyopathy, and cardiovascular mortality are more than doubled in women who have had preeclampsia.^{153,154} Severe preeclampsia, recurrent preeclampsia, preeclampsia with preterm birth, and preeclampsia with IUGR are most strongly associated with adverse cardiovascular outcomes.¹⁵³

Preeclampsia, especially in association with low neonatal birth weight, also carries an increased risk of later maternal kidney disease requiring a kidney biopsy.¹⁵⁸ Several large studies using birth and kidney registry data showed that preeclampsia increases the risk of subsequent kidney failure by almost fivefold.^{159,160} This finding has subsequently been confirmed in other populations.¹⁶¹ Familial aggregation of risk factors does not seem to explain increased ESRD risk after preeclampsia.¹⁶² Although it appears that preeclampsia is associated with increased risk of subsequent kidney failure, the absolute risk is low (e.g., 4.72 per 10,000 person-years in the Taiwanese study).¹⁶¹ Similar to cardiovascular disease, women with recurrent or preterm preeclampsia demonstrated a greater risk of future kidney disease.¹⁵⁹

Preeclampsia and cardiovascular, cerebrovascular, and kidney disease share many common risk factors, such as chronic hypertension, diabetes, obesity, kidney disease, and metabolic syndrome. Still, the increase in long-term cardiovascular mortality holds even for women who develop preeclampsia in the absence of any overt vascular risk factors. Whether these observations result from vascular damage or persistent endothelial dysfunction caused by preeclampsia or simply reflect the common risk factors shared by preeclampsia and cardiovascular disease remains speculative.¹⁶³ Regardless of etiology, it is recommended that women who experience preeclampsia, especially with preterm birth or IUGR, receive screening for potentially modifiable cardiovascular and kidney disease risk factors (hypertension, diabetes mellitus, hyperlipidemia, obesity) at their postpartum obstetrician visit and yearly thereafter. This also includes an appreciation that a 24-hour ambulatory BP assessment may be helpful in identifying women with chronic hypertension.^{164,165}

The offspring of pregnancies complicated by preeclampsia similarly have in epidemiologic studies been shown to have an increased risk of future adverse outcomes remote from pregnancy and the effects of prematurity. It remains unclear whether these adverse effects are related to environmental, developmental, genetic, or epigenetic causes.^{166,167}

SCREENING FOR PREECLAMPSIA RISK

Because it is increasingly recognized that there are preventive strategies for women developing preeclampsia subsequently in a pregnancy, a means of identifying women at risk earlier in their pregnancy becomes increasingly important. Differences in a woman's medical and family history, clinical findings, circulating biomarkers, and uterine arterial studies individually may not have adequate predictive utility, but when combined, they do hold more promise.

Different factors that can be identified in a woman's medical history, family history, and early clinical factors are significantly different in women who will go on to develop preeclampsia later in that pregnancy compared with women who will not. These have been compiled to improve the predictive accuracy in identifying women at high risk to develop preeclampsia. The strength of these risk factors has varied significantly in the literature. Individually, these risk factors have lacked the utility to be used as a screening test for the development of preeclampsia. Several professional groups (e.g., National Institute for Health and Care Excellence and the American College of Obstetrics and Gynecology) have developed simple algorithms for these clinical risk factors that have improved screening utility. In this instance, risk factors are identified as minor or major (generally based on the RR of development of preeclampsia) to improve the screening utility.¹⁶⁸⁻¹⁷⁰

Circulating markers have also been shown to differ early in pregnancy (the first trimester) in those women who will develop preeclampsia,¹⁷¹ although these markers individually have failed to have adequate predictive utility. Markers such as placental protein 13 (PP-13), PAPP-A, and angiogenic biomarkers (PIGF and sEng) have been shown to hold promise for screening of preeclampsia. PP-13 is thought to be involved in normal placentation and maternal vascular remodeling. Lower levels of PP-13 could identify women at high risk for preeclampsia as early as the first trimester.^{172,173}

Presumably as a result of failed placental vascular remodeling, preeclampsia is associated with increased placental vascular resistance and uterine artery waveform abnormalities in the second trimester, as measured by uterine artery Doppler ultrasound.¹⁷⁴ Dozens of studies have investigated the use of uterine artery Doppler for prediction of preeclampsia. Recent metaanalysis has demonstrated a pool specificity of 88% (95% CI 0.83–0.92) and sensitivity of 59% (95% CI 0.49–0.68).¹⁷⁵

Data suggest that the greatest promise may be in combining uterine artery Doppler, with serum biomarkers for preeclampsia with a woman's history (personal and family), as well as clinical findings such as BMI and BP. These have become increasingly used across the world.¹⁷⁶⁻¹⁷⁸ There are, however, limitations in access due to the requirement for expertise in undertaking these measurements and the increased cost burden of this predictive test.¹⁷⁹ Given the increased cost in the care of premature newborns, as well as the cost of treating women with hypertension,¹⁸⁰ there are increasing cost analyses demonstrating that these combined predictive tests may have a greater role in the future.¹⁸¹

PREVENTION OF PREECLAMPSIA

Many pharmacologic and nonpharmacologic interventions have been studied to prevent the development of preeclampsia in both low- and high-risk pregnancies. Preventative strategies that have been shown to be useful include antiplatelet agents, calcium supplementation, and moderate physical activity. Strategies that are currently not recommended due to a lack of consistent evidence or lack of evidence of any benefit include supplementation with long-chain fatty acid supplements,¹⁸² garlic,¹⁸³ antioxidants (vitamins C and E),^{184,185} oral magnesium,¹⁸⁶ nitric oxide,¹⁸⁷ statins,¹⁸⁸ or dietary salt restriction.¹⁸⁹ Several other agents have shown promise in animal or ex vivo studies and are still undergoing further human trials such as proton pump inhibitors¹⁹⁰ or metformin.¹⁹¹ Suggested strategies during pregnancy to reduce small and vulnerable neonatal births are further discussed in Chapter 20.

ANTIPLATELET AGENTS

Aspirin for the prevention of preeclampsia has been evaluated in dozens of trials, in both high-risk and healthy nulliparous women. It has been shown to reduce the risk of preeclampsia. The mechanism of benefit is not yet fully elucidated. Aspirin has been shown to reduce the concentration of thromboxane, reduce platelet aggregation, and inhibit vasoconstriction in uterine blood flow.¹⁹² Aspirin also has an antiinflammatory effect, mediated by aspirin-triggered lipoxins, which in turn upregulates interleukin-10 while downregulating tumor necrosis factor- α .^{193,194}

The gestation at which aspirin is commenced is significant. Aspirin that is commenced at or after 16 weeks' gestation shows minimal, if any, benefit. Comparatively, aspirin before 16 weeks significantly reduces the risk of preeclampsia by almost 50% (risk ratio 0.47, 95% CI 0.34–0.65).¹⁹⁵ Many doses of aspirin have been studied, with doses ranging from 75 to 150 mg daily.^{196,197} A range of doses is suggested (75–150 mg,¹⁶⁸ 81 mg,¹⁹⁸ or 150 mg¹⁹⁹). A meta-analysis has shown that a 150- to 162-mg dose is superior to the lower dose (75–81 mg).²⁰⁰ The timing of ingestion also affects whether aspirin prevents preeclampsia. Aspirin ingested at bedtime had greater benefit in preventing preeclampsia compared with aspirin taken in the morning.⁵⁸ Women's compliance with aspirin also affects the preventative effect of aspirin. Studies have shown that women taking <90% of doses have minimal benefit.^{201,202} Universal use of aspirin in primiparous women still remains controversial.²⁰³ Conflicting data also exist about the effect of aspirin on the prevention of preeclampsia in women with chronic hypertension. Although women with chronic hypertension were included in many of the studies in the high-risk group, one meta-analysis has shown that aspirin doesn't reduce the risk of preeclampsia but does reduce the risk of preterm birth; hence it should still be started.²⁰⁴

Clinical Relevance

Aspirin for prevention of preeclampsia should be started before 16 weeks, preferably prior to 14 weeks gestation. Aspirin 150–162 mg is superior to 75–81 mg. Evening doses are superior to morning doses.

CALCIUM

Low baseline dietary calcium intake is associated with increased preeclampsia risk.²⁰⁵ This is not well understood but may occur by low calcium stimulating parathyroid hormone or renin release, resulting in increases in intracellular calcium, which results in vasoconstriction by effecting the vascular smooth muscles.²⁰⁶ It may also benefit the placenta, by reducing uteroplacental resistance in the uterine and umbilical arteries.²⁰⁷ Many studies have been undertaken using different doses of calcium in different populations over decades. However, a meta-analysis including 18,064 women demonstrated a significant reduction in the risk of preeclampsia (RR = 0.45, 95% CI 0.31–0.65) and preterm birth (RR = 0.76, 95% CI 0.60–0.97) with high-dose (>1 g/day) calcium supplementation. The greatest effect was seen among women with low baseline calcium intake and at high preeclampsia risk.²⁰⁸ Commencing calcium supplementation before pregnancy may be beneficial in reducing adverse pregnancy outcomes (where preeclampsia is part of the composite), although this analysis was based on a single study.²⁰⁹ A previous individual patient meta-analysis has demonstrated that calcium commenced after 20 weeks may not be of benefit in preventing preeclampsia compared with the calcium being commenced before 20 weeks' gestation.²¹⁰

LOW-MOLECULAR-WEIGHT HEPARIN

Low-molecular-weight heparin is commonly used to prevent venous thromboembolism and recurrent pregnancy loss in women with antiphospholipid syndrome and inherited thrombophilias, such as factor V Leiden mutation. Several studies have evaluated the potential benefit of low-molecular-weight heparin for the prevention of preeclampsia and other placental-mediated pregnancy complications in women with a history of these complications in prior pregnancies. A meta-analysis summarized 11 of these randomized, controlled trials ($n = 1672$ women).²¹¹ The analysis (in women with recurrent pregnancy loss and persistent antiphospholipid antibodies) suggested there was no significant benefit in the live birth or preeclampsia rate. It is important to note that there was low certainty of the evidence; however, most studies had a low risk of bias in most domains.

EXERCISE

There are many physiologic benefits of exercise in pregnancy that may serve to reduce the risk of preeclampsia. Studies have shown enhanced placental growth, reduced oxidative stress, and a reduction in endothelial dysfunction.^{212–214} Regular exercise has also been shown to improve plasma volume and cardiac output and reduce inflammatory cytokines and insulin resistance.^{215,216} Recent meta-analyses have shown that interventions involving aerobic exercise resulted in a reduced risk of preeclampsia (RR = 0.63, 95% CI 0.44–0.63), GH (OR = 0.38, 95% CI 0.27–0.54), and maternal weight gain.^{199,217}

MANAGEMENT AND TREATMENT OF PREECLAMPSIA

TIMING OF DELIVERY

The timing of delivery in severe preeclampsia is controversial. Initially, women will have their BP stabilized and corticosteroids administered for fetal lung maturation where severity of the maternal and fetal condition allows. In women presenting before 24 weeks' gestation, perinatal and neonatal mortality are extremely high (>80%) even with attempts to postpone delivery, and maternal complications are common. The extent of temporizing therapies will depend on the resources available for the care of a severely preterm newborn or an unwell mother, although outcomes are improving over time.²¹⁸ In general, the presence of nonreassuring fetal testing; suspected abruptio placentae; progressive or severe thrombocytopenia; worsening liver and/or kidney function; worsening umbilical blood flow on Doppler fetal assessment; symptoms such as unremitting headache, visual changes, vomiting, or epigastric pain or pulmonary edema are generally considered indications for expedient delivery.

Generally, delivery is usually recommended in women with preeclampsia presenting before 24 weeks where expectant management is no longer adequate. By contrast, in women presenting after 37 weeks' gestation, the neonatal benefit of prolonging pregnancy is minimal and the maternal risks increase over time and outweigh the neonatal benefits. As such, delivery is indicated.²¹⁹

For women presenting between 34 and 37 weeks' gestation, the potential neonatal benefit of continuing pregnancy must be balanced against the possibility of maternal morbidity and mortality with delaying delivery. As such, expectant management is undertaken with an attempt to temporize the pregnancy. There was a lower risk of maternal mortality and severe morbidity (RR = 0.69, 95% CI 0.57–0.83), HELLP syndrome, and severe kidney impairment in women who received a planned early delivery rather than expectant management. However, there was a higher level of neonatal respiratory distress syndrome (RR = 2.24, 95% CI 1.2–4.18) and neonatal intensive care unit admission.²²⁰

Older randomized controlled trials demonstrated that in women presenting with severe preeclampsia between 28 and 34 weeks' gestation, expectant management (with delivery postponed 1–2 weeks after presentation) may result in decreased neonatal complications and decreased neonatal intensive care unit stay, with no significant increase in maternal complications.^{221,222}

There are no randomized controlled trials to evaluate optimal mode of delivery for women with preeclampsia. Furthermore, there are currently limited data to support the use of angiogenic markers in determining the timing of birth. Some data demonstrate that the rate of change of the sflt-1:PIGF ratio is greater in women who will develop delivery earlier, but there is no clear cutoff of the angiogenic ratio that is not associated with adverse fetal or maternal outcomes.¹³² Also, in women with PIGF concentrations on various point-of-care testing platforms, a low or very low PIGF concentration (<100 or <12 pg/mL) is associated with a greater risk of delivery, but it is not an indication for delivery.²²⁴

Where women are at risk of giving birth at very preterm (<34 + 6 weeks') gestation, it is important to consider the use of steroids for the benefit of the fetus. To date, there remains a paucity of data specifically in women with preeclampsia. Steroids that cross the placenta (e.g., betamethasone and dexamethasone) when administered to women at risk of preterm birth result in an improvement in neonatal morbidity and mortality. The use of steroids compared with placebo reduced perinatal death, neonatal respiratory distress, intraventricular hemorrhage, and necrotizing enterocolitis without significantly increasing maternal morbidity or mortality.²²⁵

BLOOD PRESSURE MANAGEMENT

Management of BP in preeclampsia is substantially different from that in the nonpregnant population. Rather than seeking to minimize long-term cerebrovascular and cardiovascular complications, the goal of care is to maximize the likelihood of successful delivery of a healthy infant while minimizing the chance of acute complications in the mother. Acute aggressive lowering of BP can lead to fetal distress or demise, as it may result in a reduction in placental perfusion that is already compromised, keeping in mind that the placenta cannot autoregulate vascular perfusion.²²⁶

Women with stable chronic hypertension, GH, and nonsevere preeclampsia should have their BP treated with oral agents. A number of recent RCTs^{227,228} have shown that the ideal target for BP in pregnancy in these women is ≤ 135 mm Hg and ≤ 85 mm Hg. Compared with less tight BP control, meta-analysis demonstrated that tight control of BP results in a reduction in episodes of severe hypertension (RR = 0.71, 95% CI 0.58–0.86), preeclampsia (RR = 0.85, 95% CI 0.75–0.95) and preterm birth (RR = 0.88 95% CI 0.78–0.99).¹⁹⁹

Current evidence suggests that no single antihypertensive agent is superior to another. Antihypertensives should be chosen on the basis of local availability, clinicians' familiarity, and other women's other medical issues. Some important differences are that β -blockers and calcium channel blockers are more effective than alternative agents for preventing episodes of severe hypertension.²²⁹ Details of currently available oral antihypertensives for use in pregnancy are summarized (see Table 58.5).

In women with severe hypertension (>160/110), treatment should be commenced immediately. Untreated and persistent hypertension at or above this level may result in increased hemorrhagic stroke, eclampsia, placental abruption, and adverse fetal outcomes. Most commonly, short-acting intravenous agents are used. There are, however, some oral agents that can achieve the target BP at a rate comparable with intravenous agents,²³⁰ and studies from lower-resource settings, where intravenous agents may not be available, have demonstrated success using oral antihypertensives to treat severe hypertension (Table 58.6).²³¹

MAGNESIUM FOR SEIZURE PROPHYLAXIS

Magnesium has been widely used for the management and prevention of eclampsia for decades. It has been shown to

reduce the risk of eclampsia (RR = 0.58 95% CI 0.40–0.71) in women who were considered at risk of an eclamptic seizure. Women at risk of an eclamptic seizure demonstrate features of cerebral irritability, which include hyperreflexia with sustained clonus, persistent headache, persistent visual disturbances (e.g., blurred vision or scotomata) or altered level of consciousness (e.g., confused and disoriented). Magnesium has also been shown to reduce the risk of recurrent eclamptic seizures.^{232,233} In addition, a reduction in the rate of placental abruption in the women administered magnesium sulfate for eclampsia prevention (RR = 0.67 95% CI 0.45–0.89) and reduction in a number of adverse fetal events including risk of pneumonia, ventilation, and admission to intensive care unit was also observed with magnesium therapy.²³⁴

A number of regimes have been proposed for the administration of magnesium. The standard regimen is a loading dose (usually 4 g intravenously), followed by an infusion at 1 g/hour for a total of 24 hours. This regimen has been compared with a number of modified regimens (<12-hour infusions,^{235,236} intramuscular regimens,²³⁷ or bolus-only regimens²³⁸). To date there have been little differences in outcomes between comparisons, although the quality of evidence is low and the numbers of women in the trials small. Magnesium maintenance dose should be reduced in the setting of reduced kidney function and magnesium levels monitored.²⁴⁵

MAGNESIUM FOR NEUROPROTECTION

Magnesium sulfate should be considered for women with preeclampsia to reduce the risk of eclampsia where clinically indicated. However, it is important to also consider the administration of intravenous magnesium sulfate (even where there is no indication due to the risk of eclampsia or tocolysis) for fetal neuroprotection. Individual patient data meta-analysis and subsequent RCTs have confirmed that compared with placebo, administration of intravenous magnesium sulfate reduced the incidence of cerebral palsy (RR = 0.68, 95% CI 0.54–0.86) without any effect on fetal or neonatal death.^{199,239,240}

CHRONIC HYPERTENSION

The diagnosis of chronic hypertension in pregnancy is usually based on a documented history of hypertension before pregnancy or a BP $\geq 140/90$ before 20 weeks' gestation.

The physiologic drop in BP in the second trimester, which nadirs at about 20 weeks' gestation, occurs in women with chronic hypertension and can mask the presence of underlying chronic hypertension early in pregnancy if the chronic hypertension was not recognized before pregnancy.⁵⁹ In such cases, a woman with chronic hypertension may be inappropriately diagnosed as having GH when the BP rises in the third trimester. The diagnosis of chronic hypertension is established when hypertension fails to resolve by 3 months postpartum.

Table 58.6 Agents for Use for Severe Hypertension in Pregnancy

Antihypertensives	Class of Agent	Onset of Action	Dose (Start From Low Dose and Titrate as Required)	Side Effects or Considerations
Oral nifedipine (IR)	Calcium channel blocker	30–45 minutes	10–20 mg every 30 minutes maximum of 45 mg	Headache, tachycardia, peripheral edema with recurrent usage
IV hydralazine	Vasodilator	15–20 minutes	5–10 mg every 20 minutes maximum of 30 mg	Tachycardia, headache
IV labetalol	β blocker	5 minutes	20–40 mg every 10–15 minutes, maximum of 80 mg	Bradycardia, shortness of breath Avoid in women with asthma or chronic airway disease
IV diazoxide	Vasodilator	3–5 minutes	15 mg every 5–10 minutes	Tachycardia, palpitations
Oral methyldopa ^a	α Blocker	30–120 minutes	1000 mg as a single dose	Dry mouth, sedation, abnormal liver function tests. Avoid repeated use in women with mental health issues—associated with postnatal depression
Oral labetalol ^a	β Blocker	30–120 minutes	200 mg up to 4 doses every hour	Bradycardia, shortness of breath Avoid in women with asthma or chronic airway disease

^aTried in low-resource setting. Consider where intravenous access cannot be established or is not possible.

The incidence of chronic hypertension in pregnancy is increasing over time, regardless of the age of the pregnant woman and BMI by 6% per year.²⁴¹ Chronic hypertension is more common with advanced maternal age, obesity, and black race. Pregnant women with chronic hypertension have an increased risk of preeclampsia (21%–25%), premature delivery (33%–35%), IUGR (10%–15%), placental abruption (1%–3%), and perinatal mortality (4.5%).^{242,243} However, most adverse outcomes occur in women with severe hypertension (SBP $\geq$ 160 and/or DBP $\geq$ 110 mm Hg) and those with preexisting cardiovascular or kidney disease. Both the duration and severity of hypertension, as well as the lack of systemic therapy prepregnancy, are correlated with perinatal morbidity and preeclampsia risk.¹¹⁴ The presence of baseline proteinuria increases the risk of preterm delivery and IUGR but not preeclampsia per se.²⁴⁴ The diagnosis of preeclampsia superimposed on chronic hypertension can be difficult. In the absence of underlying kidney disease, the new onset of proteinuria (>300 mg/day) or sudden worsening hypertension may indicate superimposed preeclampsia. The presence of other signs and symptoms of preeclampsia, such as headache, visual changes, epigastric pain, pulmonary edema, and laboratory derangements (e.g., thrombocytopenia, new or worsening kidney insufficiency, and elevated liver enzymes), also should prompt consideration of preeclampsia and, when present, are an indication of preeclampsia severity. Furthermore, an angiogenic ratio (sFLT1:PlGF ratio of <38) may also be reassuring that these changes are less likely to relate to evolving preeclampsia. These markers have been shown to be normal in the presence of all stages of CKD where preeclampsia is not present but the number of women in these studies was small.²⁴⁵

SECONDARY HYPERTENSION IN PREGNANCY

Secondary causes of hypertension are present in at least 10% of women with chronic hypertension in pregnancy.²⁴⁶ Women with secondary forms of hypertension may have pregnancy complication rates higher than women with primary chronic hypertension. This may be because the secondary causes were not identified during prepregnancy, and the secondary cause may be independent of the BP exacerbate poor placentation (e.g., hypoxia associated with obstructive sleep apnea). Hence prepregnancy evaluation of women with chronic hypertension should include consideration of secondary causes of hypertension including renal artery stenosis, primary hyperaldosteronism, obstructive sleep apnea (OSA), and pheochromocytoma. Unfortunately, diagnosis of these conditions during pregnancy is hampered by the fact that some of the diagnostic tests (e.g., the plasma aldosterone-to-renin ratio) are not valid in pregnancy, and other tests are relatively contraindicated due to unacceptable exposures to risk such as imaging involving radiation exposure (e.g., CT scanning, angiography, and fluoroscopy).

Renal artery stenosis due to fibromuscular dysplasia or atherosclerotic vascular disease occasionally presents in pregnancy and should be suspected when hypertension is severe and resistant to medical therapy.²⁴⁷ Diagnosis with

magnetic resonance angiography, followed by successful angioplasty in the second and third trimesters of pregnancy, has been described.²⁴⁸

Although rare, pheochromocytoma can be devastating. It has been associated with a 27-fold higher risk of adverse fetal and maternal outcomes.²⁴⁹ The outcomes are better if the diagnosis is made antenatally compared with postnatally. This syndrome is occasionally unmasked during labor and delivery when fatal hypertensive crisis can be triggered by vaginal delivery, uterine contractions, and anesthesia.²⁵⁰ Maternal and neonatal outcomes are much better when the diagnosis is made antepartum, with a multidisciplinary attentive, and aggressive medical and potentially surgical management.²⁵¹

Hypertension and hypokalemia from primary hyperaldosteronism might be expected to improve during pregnancy, as progesterone antagonizes the effect of aldosterone on the renal tubule. However, remission is not universal. Many women will develop superimposed preeclampsia.²⁵² Although there are case reports of surgical intervention during pregnancy, medical therapy remains an option. The use of spironolactone has been reported during pregnancy, but there is a risk of inadequate virilization of male fetuses due to its antiandrogenic effects.²⁵³ Eplerenone appears to be a safe alternative.^{252,254}

OSA has emerged as an important secondary cause of hypertension in pregnancy. In one study, 40% of women with hypertension in pregnancy had evidence of OSA by polysomnography.²⁵⁵ Risk factors for OSA include snoring and obesity.²⁵⁶ Recent guidelines have suggested screening for OSA should be considered in high-risk women²⁵⁷ with the best tool identified as a pregnancy-specific screening tool.²⁵⁸

APPROACH TO MANAGEMENT OF CHRONIC HYPERTENSION IN PREGNANCY

In women with chronic hypertension who are planning pregnancy, BP control should be optimized for preferably 3 to 6 months before conception using agents that are safe in pregnancy. Women should be counseled regarding the risks of chronic hypertension to the pregnancy and prevention of preeclampsia should be advised to commence at the appropriate time.

When pregnancy is unplanned, some antihypertensive medications should be stopped as soon as possible after conception (e.g., long-acting β -blockers, ACEIs, and angiotensin II–receptor blockers [ARBs]), as these may be associated with adverse fetal outcomes.^{259,260} Women should be counseled regarding the risks of adverse pregnancy outcomes including preeclampsia, preterm birth, and IUGR. Consideration should be given to the introduction measures to prevent preeclampsia (e.g., aspirin, calcium, exercise—see section on prevention). Women with chronic hypertension in pregnancy should be followed closely during pregnancy, with appropriate adjustments in antihypertensive therapy based on BP measurements and monitoring for superimposed preeclampsia.

It is important to note that even if women have normal BP during pregnancy, they may be diagnosed with chronic

hypertension soon after pregnancy. Approximately 10% of women who are normotensive during pregnancy are diagnosed with postpartum de novo hypertension within 12 months.²⁶¹

GOALS OF THERAPY IN HYPERTENSION MANAGEMENT

When hypertension is severe (SBP ≥ 160 or DBP ≥ 105 mm Hg), antihypertensive therapy is clearly indicated for the prevention of stroke and cardiovascular complications. However, it has been shown in several randomized controlled trials that women with chronic hypertension that is mild to moderate hypertension should be treated by targeting a BP of ≤ 135 mm Hg SBP and a DBP of ≤ 85 mm Hg (see “[Blood Pressure Management](#)”). Data on the timing of birth in women with stable chronic hypertension are low quality and sparse.

GESTATIONAL HYPERTENSION

Gestational hypertension (GH) is defined as the new onset of hypertension without any evidence of end organ damage (e.g., proteinuria and abnormal liver function tests) after 20 weeks’ gestation that resolves within 3 months postpartum where there was no evidence of preexisting hypertension before pregnancy. GH likely represents a mix of several underlying etiologies. A subset of women with GH has previously existing chronic hypertension that is undiagnosed. In such cases, if the woman presents for medical care during the second-trimester nadir in BP, it may be inappropriately presumed to be previously normotensive. In such a circumstance, the diagnosis of chronic hypertension is established postpartum when BP fails to return to normal.

GH can progress to overt preeclampsia. The earlier the GH is diagnosed, the greater the likelihood of preeclampsia developing.²⁶² However, even if GH is diagnosed after 36 weeks’ gestation, women have an approximately 10% chance of developing preeclampsia. Although GH was considered a milder version of preeclampsia, it is a major contributor to maternal and neonatal morbidity and mortality.²⁶³ When GH is severe, it carries similar risks for adverse outcomes similar to preeclampsia, even in the absence of proteinuria.²⁶⁴

The goals of therapy for women with GH are similar to those in women with chronic hypertension. The target for BP management is BP $< 135/85$ mm Hg. There is low quality and little data on the timing of birth in women with stable GH.

MANAGEMENT OF HYPERTENSION IN PREGNANCY

HOME-BASED BLOOD PRESSURE MONITORING

Some studies have been undertaken investigating whether women can monitor their BP at home rather than attend the hospital for extra assessments. Usually, these studies are undertaken where clinicians monitor the BPs taken at home with action plans for the women when the readings are abnormal. There have been studies undertaken with patient-based home monitoring, as well as clinician-based remote monitoring via an electronic device (e.g., mobile

phone).²⁶⁵⁻²⁶⁷ These studies have shown there is no increase in the risk of preeclampsia or GH nor other adverse maternal outcomes. Similarly, there was no difference in the gestation of delivery, rate of growth restriction, or need for neonatal admission. There was no difference in the quality-of-life scores.

It is important to note that these studies were undertaken under controlled conditions with women who were selected for inclusion in the trial using validated monitors.

CHOICE OF ANTIHYPERTENSIVE AGENTS

Recommendations for the use of antihypertensive agents in pregnancy are summarized in [Tables 58.5 and 58.6](#). It is important to note that alterations in maternal liver function and kidney function in pregnancy result in increased clearance of medications, and hence increased dosing is required compared with the nonpregnant population.^{268,269} Evidence exists that short-acting β -blockers (e.g., labetalol) and nifedipine are less likely to result in episodes of severe hypertension. With each agent, it is important to be aware of the side effects and relative contraindications of the medication being considered. For the treatment of mild to moderate hypertension in pregnancy, most agents are generally considered suitable and have no association with any fetal teratogenicity.

Agents to be avoided are long-acting β -blockers, such as atenolol, which has been associated with fetal growth restriction.²⁷⁰ Nondihydropyridine calcium channel blockers such as verapamil have been used without apparent adverse effects. However, experience with these agents is more limited than in some other classes.²⁷¹ Although diuretics are often avoided in preeclampsia, with the reasoning that circulating volume is already low, there is no evidence that diuretics are associated with teratogenicity.^{272,273} Similarly, diuretics are not considered first line in the management of chronic hypertension in pregnancy due to their theoretical impact on the normal plasma volume expansion of pregnancy. Prolonged use can result in maternal hypovolemia and increases in amniotic fluid volume²⁷⁴ above and beyond the usual side effects, such as electrolyte disturbances. When pregnancy is complicated by pulmonary edema, diuretics are appropriate and effective.

ACEIs and angiotensin receptor antagonists (ARBs) are contraindicated in pregnancy. Exposure in the second and third trimesters can lead to major fetal malformations including kidney dysgenesis, perinatal kidney failure, oligohydramnios, pulmonary hypoplasia, hypocalvaria, and IUGR.²⁵⁹ Evidence for teratogenicity with first-trimester exposure is increasing, although most studies are affected by the presence of potential confounders and ascertainment bias. A large population-based study by Cooper and colleagues²⁷⁵ reported that congenital malformations of the central nervous and cardiovascular systems were higher among women with first-trimester exposure to ACE inhibitors. A more recent study,²⁶⁰ a systematic review and meta-analysis, has similarly shown an increase in congenital malformations. Although the RR is high, the absolute risk is low and, as such, women inadvertently exposed in early pregnancy can be reassured by a normal midtrimester ultrasound examination. Less data are available on the specific effects of ARBs compared with ACEIs,²⁵⁹ and hence

it is not clearly understood if exposure to ARBs results in different adverse events compared with exposure to ACEIs.

INTRAVENOUS AGENTS FOR URGENT BLOOD PRESSURE CONTROL

Severe hypertension in pregnancy may require inpatient management, generally with intravenous agents, although based on the clinical scenario, oral agents also have a role. The agents generally studied and used in pregnancy for the treatment of severe hypertension are shown in Table 58.6. No one agent has been proven to be superior over another in terms of safety and fetal effects.²⁷⁶ Availability does vary globally, and as such it is best to use agents that clinicians are familiar with and are available locally.

Intravenous labetalol, like oral labetalol, is safe and effective, with the major drawback being its short duration of action. Some evidence exists that with a differing dosing regimen, it can also be used orally to treat severe hypertension where intravenous access is not possible or is delayed.²³¹ Intravenous nicardipine or oral nifedipine has been used safely for tocolysis during premature labor, and reports suggest that it is safe in treatment of hypertension as well.^{277,278} Increasingly, the studies comparing the effectiveness of short-acting oral nifedipine have demonstrated it to be as effective, if not more so, than some IV agents.

Hydralazine has been widely used as a first-line agent for severe hypertension in pregnancy. However, a meta-analysis of 21 trials comparing intravenous hydralazine with either labetalol or nifedipine for acute management of hypertension in pregnancy suggested an increased risk of maternal hypotension, maternal oliguria, placental abruption, and low Apgar scores with hydralazine.²⁷⁹ Intravenous diazoxide has also been studied and found to be effective in reducing BP, but it was associated with a higher rate of maternal hypotension compared with the comparator agents (labetalol or hydralazine).^{280,281} Nitroprusside carries risk of fetal cyanide poisoning if used for >4 hours and is generally avoided.

ANTIHYPERTENSIVE DRUGS IN BREASTFEEDING

There are few well-designed studies of the safety of antihypertensive medications in breastfeeding women. As a general rule of thumb, medications that have poor oral bioavailability, have low maternal serum concentrations, are highly protein bound, have large molecules, are ionized, and are not lipids are less likely to be detected in breastmilk.²⁸² In general, the agents that are considered safe during pregnancy remain so in breastfeeding.

WHITE COAT HYPERTENSION

White coat hypertension (WCH), similarly, outside of pregnancy can be diagnostically difficult to ascertain. It occurs when women have an elevated BP in the office or clinic setting and have normal BPs when they are at home. This is underdiagnosed in pregnancy but can be diagnosed with the use of a validated home monitor or ambulatory BP monitor.

The prevalence of WCH depends on the population of interest. It can be as high as 30% in women thought to have chronic hypertension in early pregnancy,²⁸³ although other studies have found a significantly lower rate (e.g., 4%).²⁸⁴ There is significant heterogeneity in the definition of WCH and how the ABPM assessment is undertaken, leading to significant variation in the results.

A metaanalysis (including women diagnosed after 20 weeks' gestation) has shown that women with WCH compared with normotensive women have a high rate of developing preeclampsia (RR = 2.35, 95% CI 1.16–4.78), as well as an increased risk of delivering a small-for-gestational-age newborn and preterm birth.²⁸⁵ However, when compared with women with chronic hypertension, women with WCH had a lower risk of preeclampsia (RR = 0.43 95%, CI 0.23–0.78). They also had lower rates of cesarean delivery and use of antihypertensive medications delivered approximately 1 week later.

MASKED HYPERTENSION

Masked hypertension (MH) occurs when a woman has normal clinic or office BP readings but abnormal BP readings on an accurate and validated home BP machine or on ABPM testing. There is a paucity of literature regarding MH in pregnancy. However, the use of home-based BP monitoring may prove to increasingly identify women with MH. The prevalence is not clear and can range from 29% in a subpopulation of high-risk women with metabolic syndrome to 6.8% in a general obstetric population.^{286,287} In both types of populations, the diagnosis of MH was associated with an increased risk of preeclampsia, cesarean delivery, small-for-gestational age neonates, and the requirement for neonatal intensive admissions compared with normotensive women.^{288,289}

PREGNANCY-RELATED ACUTE KIDNEY INJURY

DEFINITION AND CAUSES

Pregnancy-associated acute kidney injury (PR-AKI) reflects AKI occurring during pregnancy, labor, delivery, and/or postpartum. PR-AKI is a leading cause of maternal mortality and reflects the tip of the iceberg of inequitable access to CKD screening, reproductive health care, antenatal care, and potentially preventable pregnancy-related morbidity.^{290,291}

PR-AKI incidence may be underestimated due to challenges in detection (as serum creatinine is usually not routinely measured in pregnancy) and definition (as standard AKI criteria are not applicable due to physiologic changes in creatinine and GFR). PR-AKI is often defined as needing acute dialysis in pregnancy, a doubling of serum creatinine from baseline or first measure in pregnancy, oliguria, or a combination.²⁹²

PR-AKI is most commonly caused by volume depletion; reduced kidney perfusion; acute tubular or cortical necrosis (septic abortion, puerperal sepsis, hyperemesis

gravidarum or obstetric hemorrhage, placental abruption); intrarenal pathology (preeclampsia, HELLP syndrome); and medication toxicities.^{291,293} Obstructive uropathy, glomerular diseases, and multisystem conditions such as HUS/TTP and AFLP are rarer but important causes of PR-AKI. The timing of the PR-AKI can be an important clue to causation and must always be considered (Fig. 58.9).

PREGNANCY-ASSOCIATED ACUTE KIDNEY INJURY INCIDENCE, OUTCOMES, AND MANAGEMENT

PR-AKI incidence is dependent on geographic location, resource setting, and definitions used, resulting in great variability in the published literature.^{292,294-298} The PR-AKI incidence and associated mortality, especially due to septic abortion and obstetric hemorrhage, has fallen overall including in some lower-resource settings.²⁹⁹ This is driven by improved availability of safe and legal abortion, widespread antibiotic use, easy access to kidney function testing, access to good antenatal care, prompt treatment, and access to dialysis. Even so, PR-AKI has not entirely disappeared in high-income countries and may be increasing due to rises in pregnancy age and comorbidities such as diabetes, as well as pregnancies complicated by hypertension and preeclampsia.³⁰⁰⁻³⁰² In high-resource settings, most cases of AKI in pregnancy are mild and self-limited but do increase health care utilization,³⁰³ although AKI requiring dialysis is rare, affecting approximately 1:10,000 pregnancies.³⁰⁴

The PR-AKI landscape is vastly different in lower-resource settings of Asia, Africa, and South America, where it remains unacceptably high, constituting 2% to 15% of all AKI cases.^{292,298,305} Historically, PR-AKI in these settings was due to sepsis and hemorrhage, but preeclampsia is emerging as a predominant cause. Septic abortion remains a major issue where safe termination of pregnancy is illegal or not accessible. PR-AKI contributes to high rates of dialysis

requirement (up to 25% of dialysis referrals in some regions), ongoing kidney failure (3%–9% of survivors, in some regions up to 40%), and maternal mortality (ranging 3%–34%). The fetal outcomes are also comparatively worse, with high rates of fetal demise, preterm birth, and complications. The PR-AKI incidence and outcomes in these countries reflect the level of maternal and obstetric care, with many women presenting late with oliguric/anuric AKI, receiving inadequate prenatal and antenatal care, with lower awareness and ability to detect and manage PR-AKI.

The diagnostic work-up and management of PR-AKI are similar to AKI in the general population, and specific treatments are dependent on the underlying causes (see Fig. 58.9). However, gestational age, the fetal condition, and likelihood of delivery will guide decision making. For example, kidney biopsy becomes higher risk after 25 weeks (see section “Kidney Biopsy in Pregnancy”). In the third trimester, expedient timing of delivery may occur in conjunction with maternal AKI management.

Women with PR-AKI require planned follow-up to ensure kidney recovery and monitor for hypertension and CKD.³⁰⁶ However, in low-resource settings, up to a third of women do not recover baseline kidney function by time of discharge, and up to 50% will not get further kidney function testing, leaving a large cohort of women where kidney recovery may be unknown. This has important implications for the next pregnancy and future CKD and cardiovascular risk.

ACUTE TUBULAR INJURY AND CORTICAL NECROSIS

Acute tubular necrosis can be caused by any condition in pregnancy causing volume depletion, vasodilation, hypotension, and circulatory shock. Cortical necrosis is a rare, severe form of vasoconstrictive ischemic injury of uncertain pathogenesis. It is most commonly associated with pregnancy, particularly septic abortion, puerperal sepsis,

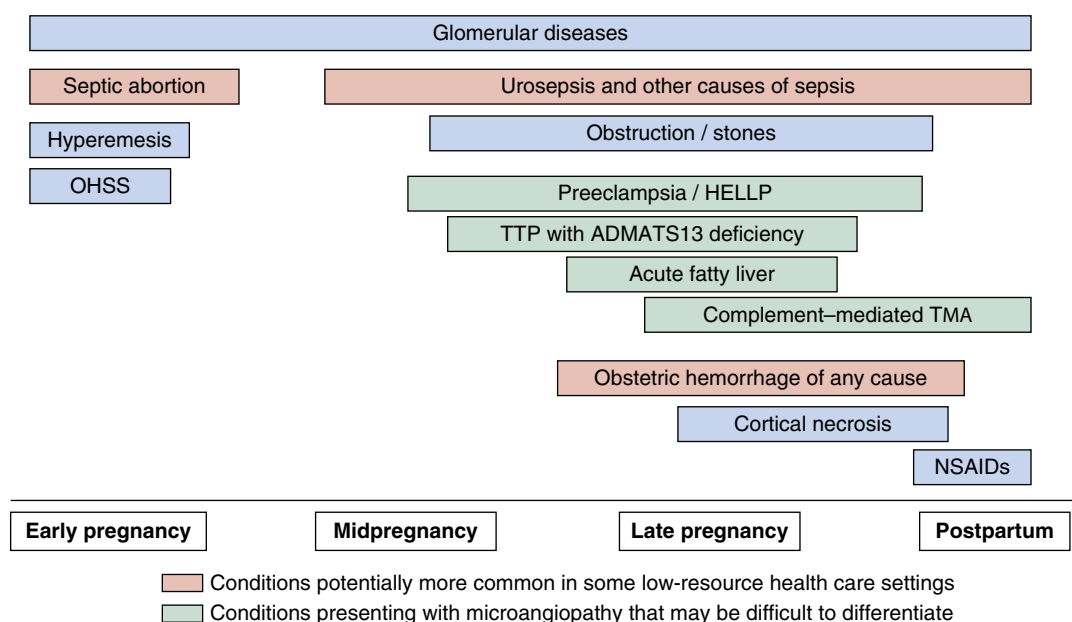


Fig. 58.9 Causes of pregnancy-associated acute kidney injury according to health setting, disease type, and gestational age. OHSS, Ovarian hyperstimulation syndrome.

eclampsia, and placental abruption. Global improvements in maternal and kidney survival have led to reduced incidence of cortical necrosis (<5% of PK-AKI),³⁰⁷ but this remains an important cause of long-term kidney failure in PR-AKI in many regions. The classical presentation is AKI with prolonged anuria. Cortical necrosis can involve the entire renal cortex (diffuse) or may be incomplete (patchy). CT imaging characteristically demonstrates a hypodense subcapsular rim. Kidney biopsy demonstrates ischemic necrosis of all cortical components.

Treatment of ATN and cortical necrosis is supportive with restoration of circulating volume, correction of underlying precipitants, consideration of expedient delivery, and commencement of dialysis. Kidney survival is usually poor in diffuse cortical necrosis even in the modern era using MRI to define disease extent.³⁰⁸ Incomplete cortical necrosis may have a protracted period of oligoanuria, followed by variable return of kidney function.

PREGNANCY-ASSOCIATED KIDNEY INJURY AND THROMBOTIC MICROANGIOPATHY

The presence of thrombotic microangiopathy (TMA) with PR-AKI may reflect certain pregnancy syndromes that share pathogenic and pathologic features, with a common final outcome of endothelial injury and microvascular thrombosis. Clinical patterns and diagnostic tests help distinguish between these entities (see [Table 58.4](#)), but in some cases the diagnosis may remain unclear. Advances in knowledge of mechanisms underlying pregnancy TMAs have led to significant improvements in clinical and therapeutic approaches.

PREECLAMPSIA

Preeclampsia is the most common cause of TMA and AKI in pregnancy (see “[Preeclampsia Epidemiology and Risk Factors](#)” earlier). In patients with severe preeclampsia and HELLP syndrome, the incidence of AKI ranges from 10% to 25% in various studies.^{309,310} Preeclampsia may promote AKI and future CKD via endothelial damage, podocyte destruction, and vascular injury.³¹¹ AKI in the setting of preeclampsia or HELLP syndrome is generally an indication for delivery.

ACUTE FATTY LIVER OF PREGNANCY

Maternal long and medium-chain fatty acid oxidation decreases in normal pregnancy, and serum fatty acid levels rise. In at-risk women, fatty acids may be hepatotoxic. Acute fatty liver of pregnancy (AFLP) is a rare but potentially fatal condition, affecting about 1 in 10,000 pregnancies, with a 2% to 10% maternal mortality rate and up to 20% fetal loss.³¹² The incidence may be increasing due to earlier detection of milder presentations. Risk factors for AFLP include multiple gestations, fetal fatty acid oxidation disorders (long-chain 3-hydroxyacyl-coenzyme A dehydrogenase deficiency³¹³), and occurrence in past pregnancies.

AFLP presents in later pregnancy with features of multiorgan fatty infiltration, predominantly with liver manifestations of elevated serum aminotransferase levels,

hyperbilirubinemia, hyperammonemia, hypoglycemia, and coagulopathy. The Swansea Criteria may be used to diagnose AFLP and predict mortality risk.^{314,315} Hemolysis and thrombocytopenia are not prominent features of AFLP (see [Table 58.4](#)). Liver biopsy is rarely performed. The resultant AKI is due to prerenal factors, plus direct fatty infiltration, hemorrhagic complications, and effects of hepatorenal syndrome.

Management of AFLP includes usual supportive care of AKI, liver failure and coagulopathy, and delivery of the fetus. Plasmapheresis has been used, but further clinical trials are required.³¹⁶ Follow-up testing for fatty acid disorders is recommended.³¹²

THROMBOTIC THROMBOCYTOPENIC PURPURA

Thrombotic thrombocytopenic purpura (TTP) is a thrombotic microangiopathic (TMA) condition caused by acquired or inherent severe deficiency of ADAMTS13 (a disintegrin and metalloproteinase with a thrombospondin type 1 motif member 13), which cleaves von Willebrand factor. TTP is characterized by ADAMTS13 activity <10%, thrombocytopenia, hemolysis, and microthrombi formation. At least half of all acute episodes of TTP are in women of childbearing age. Declining ADAMTS13 levels in the second and third trimesters may contribute to TTP risk.³¹⁷ Clinical distinction from preeclampsia/HELLP and HUS is important, and ADAMTS13 activity has aided accurate diagnosis and TTP classification in the recent era³¹⁸ (see [Table 58.4](#)). AKI is less common and severe than in other pregnancy-TMA conditions. Nearly 50% of TTP episodes in pregnancy are de novo cases, and one-third may be congenital TTP.^{318,319} Pregnancy can precipitate relapse in women with a history of TTP.³²⁰

TTP in pregnancy has high mortality (90%) if untreated. Plasma exchange with glucocorticoid therapy is beneficial in TTP but not in preeclampsia or HELLP syndrome.³²¹ Excellent fetal survival is possible with treatment,^{318,319} but intrauterine fetal death rate is high, especially for women presenting before 30 weeks' gestation.

HEMOLYTIC UREMIC SYNDROME

The landscape of pregnancy-HUS has changed due to advances in understanding complement dysregulation and genetics. Pregnancy-HUS now falls under the umbrella of complement-mediated HUS rather than secondary HUS.³²² HUS is due to a hereditary or acquired deficiency of complement regulatory proteins, leading to activation of the alternative complement pathway. Pregnancy-HUS is a rare cause of TMA and AKI, occurring usually in primiparous women, in the peripartum or early postpartum period.³²³ Kidney manifestations typically predominate, with the vast majority requiring dialysis and more than half eventually reaching end-stage kidney failure. Pregnancy-HUS may coexist with preeclampsia/HELLP, which in themselves are not clearly linked to complement activation. Therefore delivery plus management of HUS may both be required, and plasma exchange may be ineffective. More than half of women with pregnancy and postpartum HUS have been identified to have complement gene mutations, mainly CFH

mutation (30%–45%), CFI mutation (9%), and C3 mutation (9%).³²²⁻³²⁴ Complement-blockade with eculizumab has emerged as a highly promising and potentially lifesaving therapeutic intervention in many reported cases.³²⁵⁻³²⁷

KIDNEY STONES, OBSTRUCTION, AND URINARY TRACT INFECTIONS

OBSTRUCTIVE UROPATHY AND NEPHROLITHIASIS

AKI due to kidney stones and ureteral obstruction is rare, but pregnancy does increase the risk of first-time stone presentation.³²⁸ Pregnancy predisposes to calcium phosphate stones³²⁹ due to an increase in circulating levels of 1,25-dihydroxyvitamin D₃, increased intestinal calcium absorption and urinary excretion of calcium, and excessive calcium supplementation. In normal pregnancy, the concomitant increase in urine flow and physiologic dilation of the urinary tract can help protect from stone formation, and stones are also much more likely to pass.³²⁹

Ultrasonography and MRI are the preferred imaging methods.³³⁰ Because of the physiologic hydronephrosis of pregnancy, particularly on the right side, the diagnosis of true urinary tract obstruction can be challenging.

Stone management is usually conservative, with adequate hydration and analgesia. Stone-related infection should be treated aggressively, followed by antibiotic prophylaxis. Nephrostomy or stent is rarely required but is safe. Lithotripsy is relatively contraindicated during pregnancy. Stone diagnostics work-up is best timed for the postpartum period.

URINARY TRACT INFECTION AND ACUTE PYELONEPHRITIS

Urinary tract infections (UTIs) are the most frequent urinary tract problems in pregnancy.³³¹ Pregnant women are predisposed to bacteriuria and ascending pyelonephritis due to physiologic urinary status. Women with vesicoureteric reflux and polycystic kidney disease have further increased risk.³³² Dysuria and urinary frequency are common in later gestation in the absence of infection, due to pressure on the bladder from the gravid uterus; therefore confirmatory urine culture is important. Asymptomatic bacteriuria occurs in 2% to 10% of pregnancies and has increased risk of premature delivery and low birth weight.³³³ Untreated asymptomatic bacteriuria can progress to overt cystitis or acute pyelonephritis in up to 40% of patients.³³⁴ Acute pyelonephritis is a serious complication and usually presents between 20 and 28 weeks' gestation with typical symptoms or sepsis. Screening of women for asymptomatic bacteriuria in each trimester and treatment reduces pyelonephritis risk by 75 % and improves perinatal outcomes.³³⁵

Organism-targeted treatment should continue for 5 to 7 days with a follow-up culture; pyelonephritis requires aggressive parenteral therapy with drugs that penetrate the kidney parenchyma. *Escherichia coli* is the most common pathogen. Cephalosporins and penicillins are safe throughout pregnancy; trimethoprim should be avoided in the first trimester; nitrofurantoin can be used until late pregnancy; a single-treatment dose of fosfomycin upon

diagnosis is also acceptable.³³¹ Suppressive therapy with nitrofurantoin or cephalexin is recommended for those patients with bacteriuria that persists after two courses of therapy. Women with risk factors for or history of UTI should receive antibiotic prophylaxis after a single asymptomatic or symptomatic infection in pregnancy.³³⁶

CHRONIC KIDNEY DISEASE IN PREGNANCY

The true prevalence of CKD in pregnancy is difficult to ascertain, but it is estimated that CKD of any stage may affect 5% of women of childbearing age and 0.3% to 6% of pregnancies, depending on region, definitions, and data.^{337,338} These estimates come from mostly high-income countries, and the prevalence in low-income regions may be up to 20% of pregnancies.³³⁹ CKD may be rising due to an aging maternal population with comorbidities of obesity, diabetes, and hypertension. CKD may not be detected until pregnancy complications occur due to lack of routine testing of kidney function in pregnancy or younger women.³⁴⁰ Pregnancy may also unmask underlying CKD for the first time or challenge renal reserve in existing CKD.^{337,341}

Pregnancy may raise complex ethical issues, where pregnancy desire is balanced against risks to mother and baby, clinician perspectives, cultural considerations, hospital/societal level considerations of resources, optimizing utility of donated organs, and overall access to the specialized care that may be required.^{342,343} The approach to pregnancy planning and management in women with CKD or receiving KRT must integrate a myriad of ethical, sociodemographic, and clinical parameters (Fig. 58.10).

PRECONCEPTION COUNSELING

Reproductive freedom is a fundamental right, but women with CKD remain disadvantaged.³⁴⁴ Parenthood may be an important goal for many women with CKD, yet it can be fraught with feelings of grief, psychologic strain, and guilt if this aspiration remains unattainable.^{345,346} Evidence-based, overarching principles of ethical counseling include respect for reproductive autonomy and patient-centered, individualized approaches to shared decision making that reflect patient values and perspectives,^{342,343,346-348} as well as perceptions of pregnancy risk^{348,349} (Table 58.7).

Women might not always feel empowered to initiate discussions about pregnancy themselves, often relying on their clinicians to broach the subject.³⁴⁶ However, nephrologists may feel inadequately prepared to initiate these discussions.^{348,350} Counseling should be led by clinicians with expertise in pregnancy and CKD (nephrologists, obstetricians, maternal-fetal medicine specialists, obstetric physician).^{336,351} Numerous studies show that clinician support, provision of relevant information, and avoidance of negative, judgmental commentary improves the pregnancy counseling experience for women.^{336,343,345,346} Women may also value written and online information, peer support groups, and discussions with other specialized clinicians (genetics, fertility services, primary care, and psychologists).

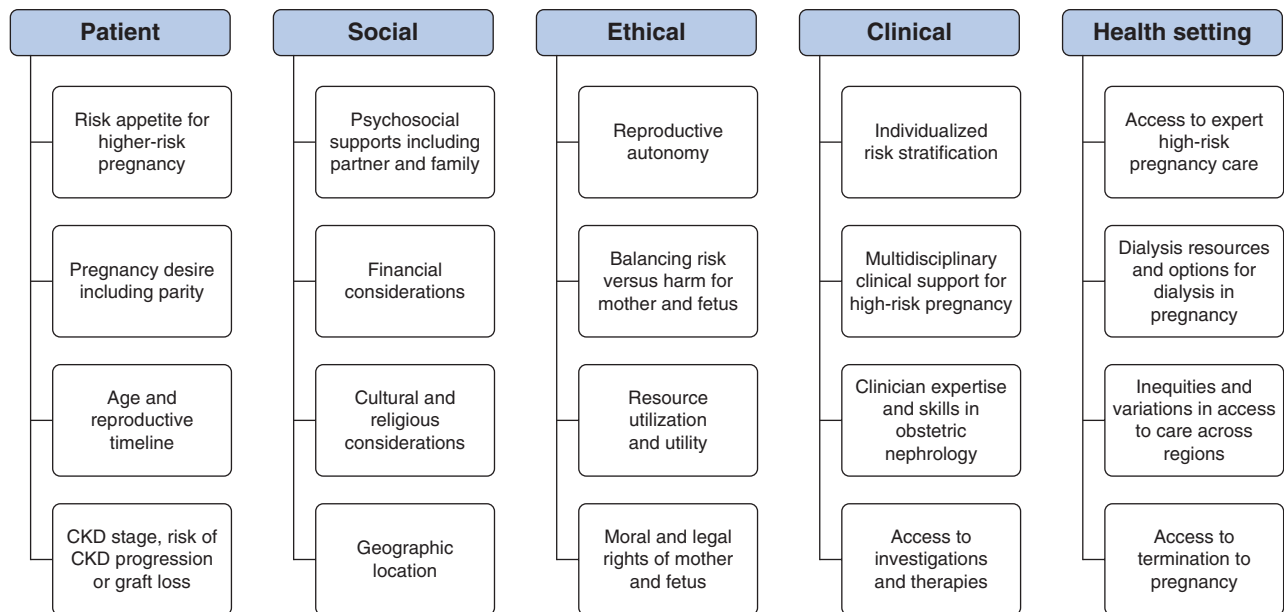


Fig 58.10 Navigating pregnancy with kidney disease: context and broad considerations that require an individualized approach tailored to each woman.

However, some women report being overwhelmed by information, so an individualized approach is essential.³⁵²

Discussions regarding reproductive options should be initiated early in the care of women with CKD and revisited periodically as the condition progresses, even if immediate pregnancy plans are not in place. These conversations serve to assist women and their health care providers in identifying the most opportune window for pursuing pregnancy, thereby averting the scenario where fertility may decline or pregnancy might exert a more pronounced impact on kidney health.

A summary of key decision points and considerations for pregnancy planning at each CKD stage is shown in Fig. 58.11, while the main elements for optimizing maternal health before pregnancy are shown in Fig. 58.12.

FERTILITY AND CONTRACEPTION IN WOMEN WITH CHRONIC KIDNEY DISEASE AND POSTTRANSPLANT

Fertility in women with CKD falls with increasing maternal age and advancing CKD stage. Uremia disrupts the hypothalamic-pituitary-gonadal axis, with blunted cyclical GnRH secretion and hyperprolactinemia. Anti-Müllerian hormone levels are reduced in women with CKD at every age. Furthermore, exposure to libido-reducing drugs and the cosmetic effects of surgery and dialysis access affect sexual drive. Women with CKD therefore have significant menstrual irregularities, anovulation, early menopause, and sexual dysfunction impacting fertility.

However, fertility may be better than anticipated, and contraception should always be considered for all women of childbearing age.³⁵³ With modern, lower-dose regimens and GnRH analogs for ovarian protection, cyclophosphamide treatment has lower impact on fertility and does not usually cause amenorrhea with 3 to 6 gm cumulative exposure. Uremia is not always contraceptive, especially in well-dialyzed

women with treated anemia.³⁵⁴ At least 30% of dialyzed women menstruate and may ovulate.³⁵⁵ Transplantation may rapidly restore fertility.

Contraception is a crucial component of navigating to successful pregnancy outcomes in women with CKD and transplants—women should be counseled that unplanned pregnancies may have worse outcomes and should be avoided.^{356,357} Contraception is particularly important where there is teratogenic medication, active primary disease or declining GFR, commencing dialysis, or immediately post-transplant, when fertility may return suddenly. Despite this, contraception remains underutilized in women with CKD, on dialysis, or post-transplant.^{354,358,359}

Nephrologists should either prescribe or refer to expert in contraceptive management and regularly confirm that contraception is implemented.³⁵⁶ Estrogen-based contraceptives may be contraindicated in many women with CKD due to hypertension, proteinuria, and thrombosis risk but may be used in normotensive women with no proteinuria and normal kidney or graft function who have no other contraindications.^{336,353} Progesterone-only oral or implantable contraceptives and intrauterine devices are generally effective and safe options for most women with CKD and post-kidney transplantation. Barrier contraception alone has a high failure rate (up to 5%), so it is not recommended. Emergency contraception strategies can be used in women with CKD as risks outweigh potential benefits of averting an unplanned pregnancy.^{336,357} Access to safe and legal termination of pregnancy is not universally available or acceptable, underscoring the need for effective reproductive counseling and contraception in women with CKD.

PREGNANCY OUTCOMES IN WOMEN WITH CHRONIC KIDNEY DISEASE

Women who enter pregnancy with CKD of any stage are at increased risk for adverse maternal and fetal outcomes. The

Table 58.7 Best-Practice Approach to Pregnancy Counseling in Women With Kidney Disease

Domain	Suggested Approach
Timing	Raise potential motherhood as early as feasible to allow planning Prospectively discuss the best window for pregnancy Allow sufficient time for evolution of discussions over multiple visits Review and revisit discussions at regular intervals
Communication	Support the woman's right to pursue pregnancy (or not) Avoid making women defend their choices Avoid judgmental comments or "forbidding" pregnancy Explain risks without catastrophizing Provide hope where possible Identify and include any other key persons (partner, family) Provide reassurance that care will be given
Patient values	Identify and acknowledge patient goals Do not assume motherhood is desired by all Acknowledge grief related to limitations to motherhood Understand how fears are balanced with desire for parenthood Define external pressures, obligations, and feelings of guilt
Decision making	Acknowledge the decisional burden Identify how much decisional control women want Assess risk based on individual clinical context Understand how risks and decisions are rationalized Determine individual appetite for "risk" Facilitate autonomy and decisional ownership Adopt shared decision making approaches
Information	Identify how much information women want to have Discuss maternal and fetal risks, long-term health impact, potential pregnancy outcomes, likely pregnancy management and progress Refer to other services (obstetric, maternal-fetal medicine, genetic, reproductive medicine) for additional information and counseling Actively facilitate and address questions

From Jesudason S, Williamson A, Huuskes B, Hewawasam E. Parenthood with kidney failure: answering questions patients ask about pregnancy. *Kidney Int Rep.* 2022;7:1477–1492.

physiologic increase in renal blood flow and hyperfiltration characteristic of normal pregnancy may place additional load on the damaged kidney, with exacerbation of hypertension and proteinuria. Multiple longitudinal cohort studies have demonstrated that increased risk for adverse pregnancy outcomes and decline in kidney function postpartum is linked to CKD stage, degree of proteinuria (especially >1 g/L), remission or activity of glomerular diseases, presence of chronic hypertension, and degree of BP control.^{341,360,361} While the live birth rate is generally excellent (>90%) even in women with kidney failure, rates of preterm birth, growth restriction and low birth weight, perinatal mortality, and neonatal intensive care unit admission are increased. Adaptive responses may be attenuated in CKD and failure of the serum creatinine >10% to drop midpregnancy is associated with worse perinatal and kidney outcomes.³⁶¹

Maternal morbidity is also more common, particularly the need for induced and operative delivery, hypertensive disorders of pregnancy, and decline in kidney function. Maternal mortality is rare in high-income settings, but kidney disease remains an important global cause of maternal death. The primary cause of kidney disease usually has less impact on outcomes than CKD stage; however, some conditions have specific problems in pregnancy. Women with nephrotic syndrome, diabetic kidney disease, and systemic lupus erythematosus (SLE) have additional risks (see section on these diseases later). Vesicoureteric reflux is associated with a higher risk of HDP and adverse outcomes.³⁵³ Women with ADPKD may be at risk of UTI and liver cyst progression in pregnancy and may be considered for preimplantation genetic diagnosis.³³² Women with IgA nephropathy generally have lower risks compared with women with other glomerular diseases but exhibit outcomes similar to other women with CKD.³⁶²⁻³⁶⁴

It is important to note that women with underlying kidney disease but only mild kidney impairment, normal BP, and no proteinuria are most likely to experience excellent maternal and fetal outcomes, with little risk for CKD progression; however, this risk is not negligible and remains higher than women without CKD.^{341,365,366} As CKD stage advances, the risks of adverse pregnancy outcomes, as described earlier, rise. In CKD stages 3 to 5, the majority of babies are likely to be born preterm—this may be driven by fetal condition, maternal comorbidity, and clinical reluctance to prolong gestation beyond 36 to 37 weeks due to diminishing fetal benefit and rising risks.³³⁶ Preeclampsia rates approach 40% in women with more advanced CKD, and more than two-thirds will have cesarean delivery. Although rates of commencing dialysis during pregnancy are <10% in women with CKD stages 3 to 5, data from the UK suggest up to 46% of women will have a sustained drop GFR of >25% or CKD stage shift or need to start dialysis after pregnancy.³⁶¹

GENERAL PRINCIPLES OF PREGNANCY MANAGEMENT

An overall approach to preconception preparation and antenatal care of women with CKD is shown in [Figs. 58.12 and 58.13](#). The principles of care are focused on optimizing maternal health, careful maternal and fetal vigilance, and

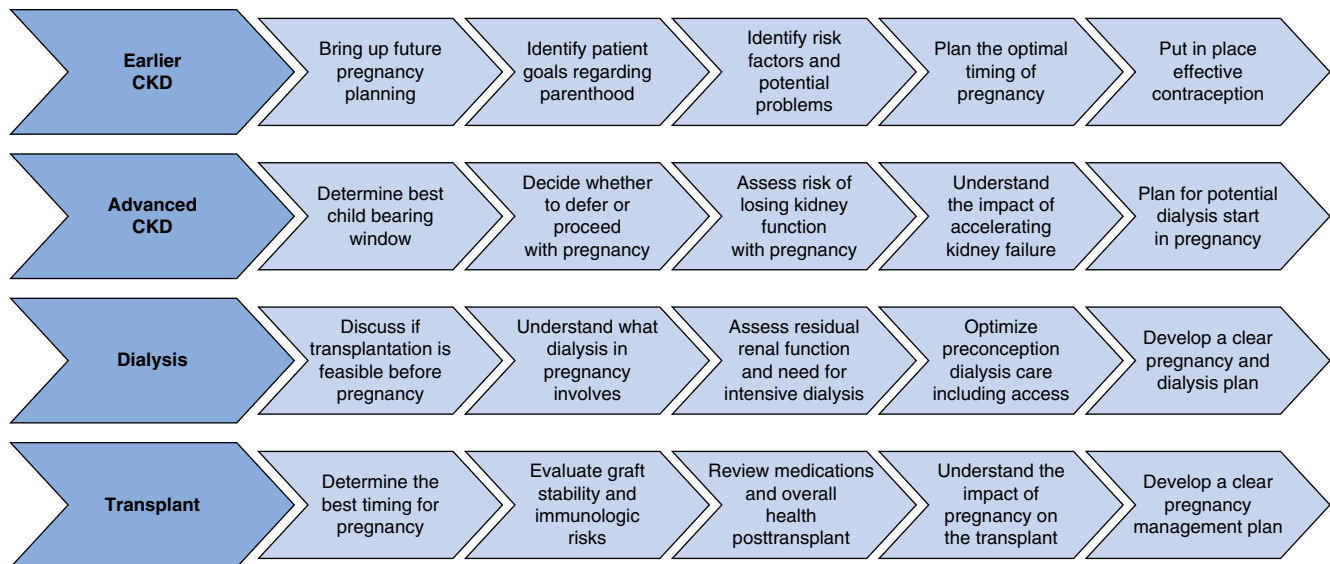


Fig. 58.11 A summary of key decision points and elements for pregnancy planning and care through the spectrum of kidney disease. (From Jesudason S, Williamson A, Huuskos B, Hewawasam E. Parenthood with kidney failure: answering questions patients ask about pregnancy. *Kidney Int Rep.* 2022;7:1477–1492.)

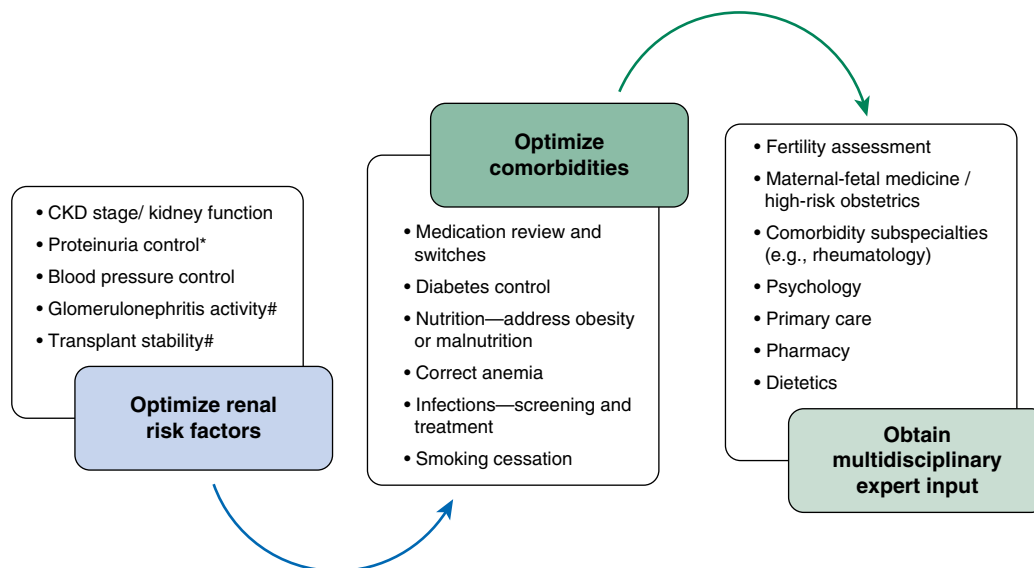


Fig. 58.12 Framework for pregnancy preparation in women with kidney disease considers continuation of renin-angiotensin-aldosterone blockade until conception in selected women where benefit outweighs risk. #Consider biopsy if uncertain.

determining the best point for delivery.^{336,353,367} Each pregnant woman should have access to a multidisciplinary team including nephrologist, obstetrician, maternal-fetal medicine, nursing, and allied health. A schedule for antenatal visits, laboratory testing, medication management, and fetal monitoring should be established. BP control (see “[Hypertensive Disorders of Pregnancy](#)” earlier for management) and aspirin for preeclampsia prophylaxis are critical elements of care. Women may need to relocate from rural areas to access specialized care. In many countries, there may be a financial burden associated with high-risk pregnancy. Emotional and psychological support should also be considered.

SPECIFIC CONSIDERATIONS IN PREGNANCY WITH KIDNEY DISEASE

KIDNEY BIOPSY IN PREGNANCY

Kidney biopsy as part of the *preconception* work-up may be highly informative regarding diagnosis, disease activity, and fibrosis indicative of renal reserve. Kidney biopsy may be indicated *during* pregnancy where there is clinical suspicion of glomerular disease or transplant rejection. Biopsy is warranted only where the disease is severe enough to impact pregnancy and the establishment of a pathologic diagnosis is likely to change management *within* the time frame of the

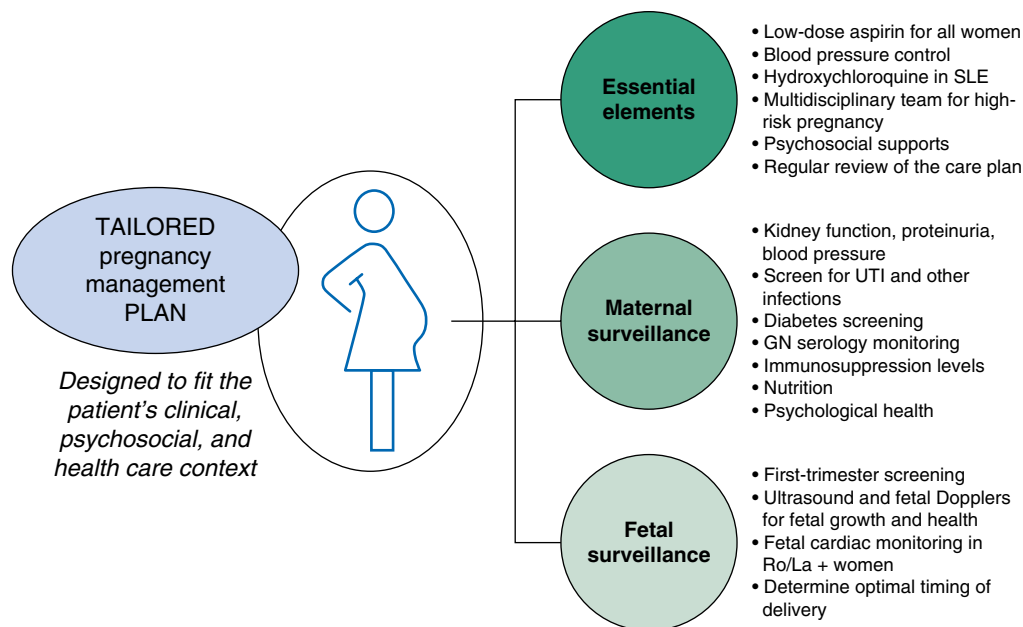


Fig. 58.13 A framework for antenatal care in women with kidney disease. This care plan should be determined preconception and revised at conception and regularly as the pregnancy progresses. The frequency of monitoring and visits will be individualized according to chronic kidney disease stage and pregnancy risk. *SLE*, Systemic lupus erythematosus; *UTI*, urinary tract infection.

pregnancy.³⁶⁸ Where delivery is likely before treatment can take effect, it is better to deliver and then biopsy. Many studies on kidney biopsy in pregnancy were from the preultrasound era. A 2013 review of 243 kidney biopsies performed during pregnancy and 1236 kidney biopsies performed during the postpartum period³⁶⁹ revealed kidney biopsies performed during pregnancy had a higher complication rate than biopsies performed after pregnancy (7% vs. 1%). Only four cases of major bleeding occurred, all in women who were biopsied between 23 and 26 weeks' gestation. This led to recommendations to avoid kidney biopsy after 23 to 26 weeks' gestation. More recent series have revealed a low rate of complications but high rate of diagnosis and change of management in pregnancy following biopsy up to 30 weeks.³⁷⁰⁻³⁷² Transplant biopsy may be lower risk due to location and may have greater risk-to-benefit ratio, where there may be risk of graft loss.³⁷³ Overall, biopsy should be used judiciously in pregnancy. Follow-up is essential: Postpartum biopsy usually has a high diagnostic yield.³⁷⁴

NEPHROTIC SYNDROME IN PREGNANCY

Proteinuria is more likely to commence or worsen in pregnancy as CKD stage advances or where glomerular disease is not controlled. Serum albumin falls in normal pregnancy due to dilution, but in a proteinuric woman, low serum albumin (<28 g/L) may reflect nephrotic syndrome. Hyperlipidemia and edema are less discriminatory as these may be observed in normal pregnancy. The usual effects of urinary protein loss are observed in nephrotic pregnant women including loss of transferrin (iron deficiency), immunoglobulins (infection), and antithrombin III (venous thromboembolism).

The management of primary glomerular diseases causing nephrotic syndrome is altered in pregnancy, with a focus on steroids, azathioprine, and calcineurin inhibitor

(CNI), and potentially rituximab as third-line therapy.³⁶⁸ BP control is essential. Substantial edema may be managed with diuretics, with monitoring for oligohydramnios and placental perfusion. Venous thromboembolism prophylaxis is recommended,³³⁶ given the prothrombotic risks of nephrotic syndrome and pregnancy, especially with other risk factors (obesity, immobility). Low-molecular-weight heparin dosing should be adjusted for renal function and should continue for 6 weeks postpartum. RAAS blockade should be commenced postpartum. Women with no clear diagnosis should be followed up postpartum and biopsied if proteinuria does not resolve within 6 months. Postpartum biopsy has a high diagnostic yield.³⁷⁴ Where there may be a genetic basis for nephrotic syndrome (e.g., focal segmental glomerulosclerosis), genetic testing may be considered for mother and baby.

MEDICATION MANAGEMENT IN WOMEN WITH CHRONIC KIDNEY DISEASE, GLOMERULAR DISEASES, AND TRANSPLANT

MANAGING RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM BLOCKADE PRECONCEPTION

ACEIs and ARBs are contraindicated in the second and third trimesters of pregnancy, leading to birth defects including hypocalvaria (hypoplasia of the membrane bones of the skull), renal tubular dysplasia, and IUGR.³⁷⁵ ARBs may carry a higher risk.³⁷⁶ Multiple historical and more recent cohort studies have found no adverse fetal effects with first-trimester ACEI and ARB exposure above that of other antihypertensives and hypertension itself.^{377,378} Women with a strong indication for RAAS blockade, such as proteinuric GN or diabetic nephropathy, may be continued on these medications while attempting conception and discontinue as soon as pregnancy is recognized. This avoids prolonged time without the nephroprotective effect

of these drugs while waiting to conceive (which may take years). The alternative approach of ceasing ACEI or ARB and assessing pregnancy risk based on proteinuria while off these agents may lead to overtreatment of proteinuria with additional biopsies and immunosuppression. Overall, an individualized approach is recommended according to patient factors and preferences and indication for RAAS blockade.^{336,368}

IMMUNOSUPPRESSION IN PREGNANCY

The potential effects and recommendations for common immunosuppressive drugs are shown in detail in [Table 58.8](#). In summary, CNIs (either cyclosporine or tacrolimus), glucocorticoids, and azathioprine are the “safest” options for immunosuppression during pregnancy. Prednisone stress dosing may be required during acute illness and labor; higher doses may be used to treat rejection or acute glomerular disease. CNI dosing may be challenging in pregnancy due to decreased gastrointestinal absorption, increased volume of distribution, albumin and hematocrit dilution, increased renal clearance, and alterations in unbound (active) fraction in whole blood.³⁷⁹⁻³⁸¹ On the basis of whole blood levels, dosages of CNIs generally need to be increased in pregnancy to maintain therapeutic range^{382,383}; however, this may lead to excessive exposure—close monitoring of drug levels and signs of toxicity is essential.

TIMING AND MODE OF DELIVERY

The timing of delivery will be determined by maternal and fetal condition, with a goal for term delivery (over 37 weeks), if the fetal gains of longer gestation continue to outweigh risks of prolonging pregnancy. However, in many women with CKD, preterm delivery will be necessitated due to preeclampsia, worsening maternal BP or kidney function, or declining fetal growth or condition. Therefore timing of delivery should be under constant review. Isolated serum creatinine rises toward the end of pregnancy should not be the sole indication for delivery, as this also occurs in normal pregnant cohorts.¹⁴ CKD or kidney transplant itself does not mandate cesarean delivery; this should occur for obstetric indications only. Vaginal delivery is safe and may be preferred. Transplant anatomy should be defined well ahead of delivery by the obstetric team, with involvement of the transplant surgical team if required.³⁴⁷

There should be vigilance intrapartum and postpartum regarding fluid balance, urine output, blood loss, and use of nephrotoxic analgesia (NSAIDs) and antibiotics. Urgent postdelivery transplant imaging for any reduction in function or urine output.³⁴⁷ Women will require close follow-up in the immediate and short-term postpartum periods.

BREASTFEEDING AND DRUGS

Breastfeeding should be encouraged in women who wish to do so. Most pregnancy-safe medications are usually also safe in lactation. The Drugs and Lactation Database (LactMed)³⁸⁴ is an excellent online resource. Studies on transfer of CNIs to the babies of breastfeeding mothers are inconsistent but overall suggest breast milk exposure

Table 58.8 Immunosuppressive Medications in Pregnancy

Drug	Recommendations
Prednisone	Safe when used long term at low to moderate doses (5-10 mg/day) Safe when given acutely at high doses for treatment of acute rejection or induction therapy for glomerular disease
Cyclosporine, tacrolimus	Extensive clinical data suggest safe at low-to-moderate clinical doses Changes in absorption and metabolism require close monitoring of levels and frequent dose adjustments in pregnancy; however, therapeutic range may need to be lower to avoid toxicity from active drugs. The active (unbound) fraction rises in response to physiologic changes in hematocrit and albumin in pregnancy. Whole blood levels need careful interpretation.
Azathioprine	Considered safe at dosages <2 mg/kg per day, but at higher doses associated with fetal growth restriction
Mycophenolate mofetil	Contraindicated—teratogenic in animal and human studies, associated with early pregnancy loss and fetal defects
Sirolimus, everolimus	Contraindicated—embryo/fetal toxicity in rodents; human studies are lacking beyond case reports/ anecdotal data
Rituximab (anti-CD20 mAb)	Use in pregnancy is growing, no clear signal of adverse fetal effects. May be used before conception or during pregnancy where benefit outweighs potential risks
Muromonab-CD3 (OKT-3)	Case reports of successful use for induction in unsuspected pregnancy and for acute rejection, but data are limited
Antithymocyte globulin (C)	Avoid in pregnancy; Very limited animal and human data
Belatacept, basiliximab, alemtuzumab	Avoid in pregnancy; inadequate safety data in humans

Drawn from selected references.^{336,347,368,380,381,409,437,456-472}

is much lower than in utero exposure.³⁸⁵⁻³⁸⁷ Monitoring neonatal blood concentrations is not usually necessary. Theoretically, MMF should be safe in breastfeeding because the active metabolite secreted in breast milk, MMA, is not gastrointestinal bioavailable; however, human evidence of safety is limited to a few case reports. There are also insufficient data regarding safe breastfeeding with angiotensin receptor antagonists, statins, SGLT2 inhibitors, fineronone, and mTOR inhibitors. Enalapril and quinapril can be used to reinstate RAAS blockade in proteinuric women.³⁸⁴

DIABETIC KIDNEY DISEASE AND PREGNANCY

Diabetes prevalence in pregnancy remains <1% but has doubled from 1990 to 2020.³⁸⁸ Up to 50% of cases are due to type 2 diabetes. The prevalence of diabetic nephropathy is less certain. A large population study from Canada revealed diabetic nephropathy occurred in 1.4% of nearly 87,000 women with diabetes in pregnancy.³⁸⁹ Women with diabetes receiving kidney replacement therapy have lower birth rates,^{390,391} possibly due to lower fertility and less encouragement toward pregnancy. Women with pregestational diabetes without nephropathy have a higher risk of fetal adversity including congenital malformation, preterm birth, macrosomia, birth trauma and perinatal mortality, and maternal adversity of preeclampsia, acute kidney injury, and hospitalization.^{303,392-394} These risks are exacerbated in women with nephropathy, with 2.3- to 4.2-fold increased odds of preeclampsia or chronic hypertension with superimposed preeclampsia.³⁸⁹ The presence of even microalbuminuria is predictive of preeclampsia risk and preterm delivery.³⁹⁵ The risk of decline in kidney function following pregnancy in women with diabetic nephropathy has been poorly studied but is likely dependent on the baseline kidney function, magnitude of proteinuria, and glycemic and BP control. Poor glycemic control before and during pregnancy also substantially worsens outcomes.^{396,397}

Overall, women with CKD and diabetes are a particularly high-risk cohort who warrant specific counseling and vigilance in pregnancy. Guidelines universally recommend that attaining glycemic control preconception without risking hypoglycemia is an important strategy to improve outcomes.^{398,399} Exercise and minimizing weight gain preconception and antenatally are useful adjunctive strategies.³⁹⁹

Diabetes may also subsequently affect metabolism and kidney function in offspring^{400,401} with reduced nephron mass and create higher risks for future kidney disease and hypertension.

LUPUS NEPHRITIS AND PREGNANCY

SLE is most common in women of childbearing age. Active SLE and treatments for SLE (cyclophosphamide) may have an impact on ovarian reserve and fertility. Women with SLE are at increased risk for preterm birth, IUGR, spontaneous abortions, and preeclampsia.^{402,403} Outcomes for lupus are poorer than for women with comparable kidney functional impairment from other causes, particularly if associated with antiphospholipid antibodies or lupus anticoagulant.^{404,405} Risk factors for adverse pregnancy outcomes include lupus disease activity preconception or a flare in pregnancy, lupus nephritis, antihypertensive use, glucocorticoid use (reflecting disease severity), antiphospholipid antibodies, thrombocytopenia, and nonwhite or Hispanic race.^{402,403,406,407} Discontinuation of hydroxychloroquine in pregnancy is associated with an increased risk of lupus flare, so this agent should be continued in all women with SLE, particularly nephritis.⁴⁰⁸ Women should be counseled that outcomes will be better if pregnancy-safe immunosuppression is in place and stabilized at maintenance levels for at least 3 months and overall lupus activity is quiescent for ~6 to 12 months.^{368,409}

Approximately 10% to 15% of women with a history of lupus nephritis experience a nephritis flare during pregnancy.^{404,410} Differentiating lupus nephritis flare and preeclampsia can be a clinical challenge due to overlapping features of proteinuria, hypertension, and thrombotic microangiopathy. Low C4, but not low C3 or anti-double-stranded DNA antibody positivity, is predictive of pregnancy-associated lupus nephritis flare and complement changes can herald a flare.⁴¹⁰ An active urinary sediment suggests lupus nephritis. In some cases, kidney biopsy may be indicated (see the “[Kidney Biopsy in Pregnancy](#)” section). Angiogenic biomarkers sFLT1 and PlGF are useful adjunct tests.⁴¹¹ De novo SLE in pregnancy should be worked up similarly to the nonpregnant state, with kidney biopsy carefully considered as discussed earlier. The presence of SSA(Ro) or anti-SSB(La) is associated with neonatal lupus in 1% to 2% and fetal congenital heart block.

All women with lupus nephritis should receive hydroxychloroquine to reduce risk of adverse pregnancy outcomes and congenital heart block,⁴¹² aspirin for preeclampsia risk reduction, and thromboprophylaxis if they are nephrotic or have antiphospholipid antibodies. Steroids, azathioprine, and calcineurin inhibitors are first- and second-line agents.^{368,409} Rituximab, mycophenolate, or cyclophosphamide may rarely be considered as rescue therapy in severe and life-threatening disease; however, termination of pregnancy (therapeutic abortion or delivery) is highly likely to be required in this setting. The safety of newer agents such as belimumab in pregnancy has not been established and is not recommended.

PREGNANCY IN WOMEN RECEIVING KIDNEY REPLACEMENT THERAPY

Pregnancy outcomes in dialyzed and transplanted women are summarized in [Fig. 58.14](#), while [Table 58.9](#) provides an overall framework for the antenatal management of this cohort.

PREGNANCY IN WOMEN RECEIVING DIALYSIS

Kidney failure is characterized by severe hypothalamic-pituitary-gonadal dysfunction, which is reversed most successfully by transplantation, but it may also be improved with effective correction of anemia and uremia. While women of childbearing age receiving dialysis frequently have menstrual disturbances, anovulation, and infertility, pregnancy is possible and pregnancy rates are rising.^{390,413} The vast majority of pregnancies are in women receiving hemodialysis; birth rates in peritoneal dialysis recipients are much lower as women are often switched to hemodialysis, and peritoneal dialysis itself may impede ovulation and fertility.⁴¹⁴ Contraception uptake in dialyzed women is low³⁵⁴ but should not be overlooked.

In the modern era in high-income settings, the live birth rate in dialyzed women has significantly improved in some centers and may be similar to transplanted women. Maternal mortality is negligible, although maternal and fetal complications are substantially increased⁴¹⁵ (summarized in [Fig. 58.14](#)). In lower-resource settings, late presentation of pregnant women with advanced kidney failure needing

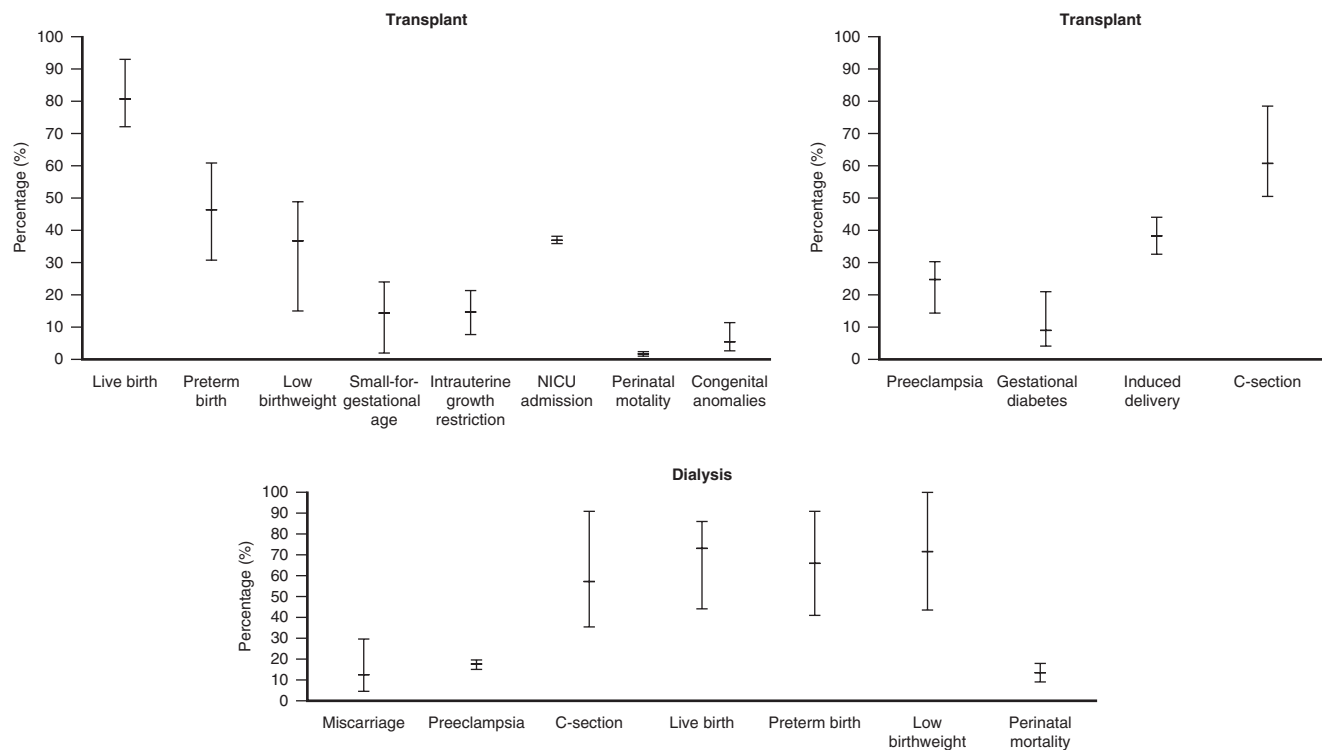


Fig. 58.14 A summary of the range of key maternal and fetal outcomes reported for pregnancies in women with a kidney transplant or receiving dialysis, based on studies large cohort studies drawn from post-2000 contemporaneous cohorts in developed countries.^{413,419,420,429,431,442,447-455} Small-for-gestational age is defined as birth weight <10th percentile for gestational age. *NICU*, Neonatal intensive care unit. (Adapted from Jesudason S, Williamson A, Huuskos B, Hewawasam E. Parenthood with kidney failure: answering questions patients ask about pregnancy. *Kidney Int Rep.* 2022;7:1477-1492.)

dialysis and pregnancy in chronically dialyzed women continue to have poorer outcomes.⁴¹⁶

Understanding that effective control of uremia was a crucial determinant of pregnancy outcome⁴¹⁷ led to a shift toward longer hours of dialysis and subsequent gains in fetal survival and maternal outcome.⁴¹⁸ In women with little residual kidney function and established on dialysis, intensive hemodialysis (>36 hours/week) is associated with gains in live birth rate, gestational age, and fetal birth weight; less polyhydramnios; and superior hypertension control but requires daily or nocturnal dialysis to deliver the hours.⁴¹⁹ Intensive hemodialysis is burdensome to the patient and requires enormous resource commitment and may not be feasible to implement in many settings. Home hemodialysis to deliver extended hours of dialysis also poses challenges to the patient and clinical team, and this model of care is possible only in some centers.⁴¹⁴

In women with residual kidney function, shorter hours of dialysis in pregnancy may suffice. Successful pregnancy is higher in women who conceive before initiation of dialysis, reflecting the importance of residual kidney function.⁴²⁰⁻⁴²² Dialysis dose should therefore be individualized to the woman, adjusted as pregnancy progresses, and titrated to predialysis urea <12.5 mmol/L.^{347,414} The optimal time to start dialysis in pregnancy has little evidence and wide practice variation; expert consensus suggests commencing dialysis at urea 17 mmol/L; however, the clinical context and gestational age should be considered.^{336,423} Creation of arteriovenous fistula may help minimize catheter use, but both access options for commencing dialysis are suitable.

Data on pregnancy outcomes in women on peritoneal dialysis (PD) are limited to case reports and small case series, but successful pregnancies are now well reported.^{414,415} Complications may include peritonitis, catheter malfunction, exit site infection, mechanical pressure from the gravid uterus, and poor clearances.⁴²⁴ PD may be associated with a higher rate of small-for-gestational-age infants, as compared with hemodialysis.⁴¹⁸ Women already on PD can potentially stay on PD for some of the pregnancy, especially where there are constraints on hemodialysis. PD is rarely commenced in pregnancy but has the advantage of less hemodynamic challenge compared with hemodialysis, stable metabolic control, and preservation of residual function.⁴²⁵ Hybrid treatment or switching fully to hemodialysis is often required to maintain comfort and clearances.⁴²⁵⁻⁴²⁷

PREGNANCY IN THE KIDNEY TRANSPLANT RECIPIENT

BIRTH RATES IN KIDNEY TRANSPLANT RECIPIENTS

Kidney transplantation may restore normal hormonal function and fertility within 6 months in approximately 90% of women of childbearing age.⁴²⁸ Contraception is likely underutilized in this cohort³⁵⁸ but must be part of standard care peritransplant. Birth rates in transplanted women have risen since the 1990s but may be plateauing in the recent era in some countries.^{390,391,413,429} Widespread use of mycophenolate immunosuppression from the late 1990s may have had an impact on lower reported birth rates post-transplant, but this remains speculative. Transplanted

Table 58.9 Recommended Medical Management for Pregnant Women With Kidney Failure

Dialysis Recipient	Transplant Recipient
Model of Care	
- Establish a team including nephrologists, obstetricians with maternal-fetal medicine and high-risk pregnancy experience, midwives and allied health support including dietetics, psychology, and pharmacy. Identify the coordinator of care. - Care close to a tertiary center for expert nephrology, transplant, and obstetric care	
Medications	
- Stop phosphate binders, calcimimetics - Switch to pregnancy-safe antihypertensives - Commence low-dose aspirin 75-150 mg daily	- Switch to pregnancy-safe antihypertensives and immunosuppression - Cease antiviral therapy - Commence low-dose aspirin 75-150 mg daily
Modality-Specific Management	
- Measure residual function - Consider PD switch to HD where feasible - Titrate dialysis hours and frequency based on residual function; aim for predialysis urea 12 mmol/L - Implement intensive dialysis regimen (>36 hours/week, titrated to predialysis urea) in women with no residual function - Do not use Kt/V to assess clearances - Use minimal anticoagulation - Regularly review and titrate dialysate composition based on bicarbonate, potassium, calcium, and phosphate blood levels - Adjust peritoneal dialysis prescription to increase clearances; reduce volume if required for patient comfort - Frequently review and adjust dry weight by 0.3-0.5 kg/week from second trimester - Monitor dialysis access closely for infection and complications - Minimum weekly clinical review	- Minimum monthly clinical review and measurement of kidney function, proteinuria, and immunosuppressive trough levels - Increase frequency of review as gestation progresses - Anticipate a fall in whole blood (bound) trough CNI levels in the second trimester. Interpret falling CNI levels cautiously and beware of potential toxicity after dose increases - Consider transplant biopsy if acute graft dysfunction without clinical suspicion of preeclampsia; balance value of biopsy against gestational age and likelihood of early delivery
Blood Pressure	
- Recommend home BP monitor - Vigilance for superimposed preeclampsia - Consider sFLT1/PIGF levels where available - Aim for BP <135/85 - Avoid placental hypoperfusion from intradialytic hypotension	- Recommend home BP monitor - Vigilance for superimposed preeclampsia - Value of sFLT1/PIGF levels uncertain - Aim for BP <135/85 - Avoid hypotension <110/70
Anemia	
- Use ESA to achieve target Hb 110 g/dL - Ensure iron stored is replete according to local protocols - Develop criteria for blood transfusion—optimize hemoglobin to avoid transfusion	
Infection	
- Monitor closely for catheter sepsis or AVF infection - Screen for and treat bacteriuria/UTI; consider prophylaxis - Coronavirus vaccination	- Screen for and treat bacteriuria/UTI; consider prophylaxis - Coronavirus vaccination - Screen/Monitor for CMV in women at risk (past CMV, previous prophylaxis)
Nutrition	
- Start or increase dose of vitamins B and C and folic acid supplements - Reduce fluid, potassium and phosphate restrictions; may require supplementation - Expert dietitian involvement to optimize protein and calorie intake	Dietitian review as required
Diabetes Screening	
- Early screening (16-20 weeks) in high-risk patients (prednisolone, tacrolimus, family history, past gestational diabetes mellitus, obesity) - Routine screening at 28 weeks' gestation or per local practice	
Diabetes Management	
- Diabetes education and dietitian - Insulin therapy	- Diabetes education and dietitian - Metformin or insulin

Continued on following page

Table 58.9 Recommended Medical Management for Pregnant Women With Kidney Failure (Continued)

Dialysis Recipient	Transplant Recipient
Fetal Monitoring	
- Dating, morphology scans, and first-trimester screening as per local practice - Consider noninvasive prenatal testing (cell-free DNA) for aneuploidy - Frequent (fortnightly) growth scans in third trimester to monitor growth restriction, amniotic fluid index, and vascular changes suggesting preeclampsia	
Delivery and Early Postpartum Care	
- Stop aspirin at 35 weeks' gestation - Develop and regularly review the “goalposts” for delivery including timing - Trial of labor is not contraindicated - Clarify transplant anatomy before delivery - Steroid “stress dose” if on chronic glucocorticoid therapy - Avoid nonsteroidal antiinflammatory medications for pain relief - Consider venous thromboembolism prophylaxis following operative delivery - Urgent postdelivery transplant imaging for any reduction in function or urine output - Establish breastfeeding if desired by mother; support decisions about breastfeeding	
Postpregnancy Care	
- Contraception in place - Psychosocial support - Resume nonpregnant dialysis prescription - Monitor residual kidney function	- Contraception in place - Psychosocial support - Close monitoring of graft function and CNI levels - Adjust/reinstate medications

AVF, Arteriovenous fistula; BP, blood pressure; CMV, cytomegalovirus; CNI, calcineurin inhibitor; PIGF, placental growth factor; sFLT1, soluble fms-like tyrosine kinase 1; UTI, urinary tract infection.

Reproduced from Jesudason S, Williamson A, Huuskos B, Hewawasam E. Parenthood with kidney failure: answering questions patients ask about pregnancy. *Kidney Int Rep.* 2022;7:1477–1492.

women with diabetes have the lowest pregnancy and birth rates, which may reflect specific counseling against pregnancy due to perceived risk.

TIMING OF PREGNANCY AFTER TRANSPLANTATION

Pregnancy within the first 6 to 12 months following transplantation is undesirable due to acute rejection, immunosuppression, and infection risk. Women should be advised to wait 1 to 2 years after transplant.^{336,430} The American Society of Transplantation suggests that conception could be safely considered as early as 1 year post-transplant for women on stable doses of nonteratogenic immunosuppressive agents, with normal kidney function (Cr <1.5 mg/dL and urine protein excretion of <500 mg/day), no concurrent fetotoxic infections (especially cytomegalovirus), and no rejection episodes within the past year.

OBSTETRIC AND PERINATAL IN KIDNEY TRANSPLANTED MOTHERS

Outcomes for pregnancies in transplanted women in the modern era are summarized in Fig. 58.14. Rates of early pregnancy loss (either miscarriage or termination of pregnancy) in transplanted women range from 6% to 25%.^{429,431} Overall, live birth rates are 72% to 93% (depending on whether early pregnancy loss is excluded or not). Pregnancies in women with transplants remain high risk, even with good graft function.^{431,432} A major driver of adverse pregnancy outcomes is preeclampsia, which

historically has been observed in up to 37% of women, and in Australia it been reported as high as 48%⁴³³ in the tacrolimus era. This may reflect better awareness and diagnosis of preeclampsia or graft dysfunction related to physiologic changes in CNI exposure and measured drug levels in pregnancy (see “Immunosuppression” section). Placental histology is commonly abnormal reflecting maternal vascular malformation in transplanted women exposed to CNI.⁴³⁴ Risk factors of chronic hypertension, previous preeclampsia, or preconception serum creatinine >125 mmol/L have cumulative effect, and women with all three factors have a 96% risk of preeclampsia.⁴³⁵ Gestational diabetes is not consistently increased in studies of transplanted women.⁴²⁹

Cesarean delivery occurs in up to 50% of transplanted women, possibly driven by comorbidities, preeclampsia, obstetric indications, or patient/clinician choice. A trial of labor may be safer than scheduled operative delivery in terms of neonatal outcomes.⁴³⁶

Preterm delivery <37 weeks occurs in 30% to 60%, preterm birth <34 weeks in up to 25%, and low birth weight in 15% to 50% of babies (see Fig 58.14). Structural birth defects are similar to the general population.⁴³⁷ Studies following up babies of transplanted mothers are limited, but long-term impacts are not clearly evident.⁴³⁸

EFFECT OF PREGNANCY ON KIDNEY ALLOGRAFT FUNCTION AND SURVIVAL

Neither pregnancy itself nor the occurrence of preeclampsia clearly adversely impacts long-term graft survival or function

in stable women with good graft function; preconception GFR is the major determinant of graft outcome after pregnancy.^{433,435,439-442} These studies have many unmeasured confounders including proteinuria, BP, donor-specific antibodies, and immunosuppression exposures. As with CKD, the degree of kidney functional impairment and comorbidity determines pregnancy and graft outcomes.

ACUTE REJECTION IN PREGNANCY

Acute rejection during pregnancy is uncommon (1%–10%) and may be difficult to differentiate from preeclampsia or CNI toxicity;⁴⁴¹ therefore biopsy should be performed if gestation will be continued. High-dose steroids, intravenous immunoglobulin, and optimization of CNI are the mainstay of treatment. Because other rescue therapies are not safe (see Table 58.8 on immunosuppression), refractory rejection usually warrants delivery and then definitive therapy.

PREGNANCY OUTCOME FOLLOWING KIDNEY DONATION

The RRs of GH and preeclampsia are doubled in living kidney donors as compared with matched nondonors, but the absolute risk remains acceptably small (<10%) and there are no clear adverse fetal sequelae.⁴⁴³⁻⁴⁴⁵ These data are drawn mostly from white women in high-income settings, and therefore broad applicability should be tempered. Guidelines state potential donors of reproductive age should be carefully and individually counseled; however, future pregnancy alone is not a reason to deny donation.⁴⁴⁶

CONCLUSION

Recent decades have brought significant advances in our understanding of pregnancy in women with CKD and glomerular disease and who were posttransplant. The paradigm has shifted to shared decision making and greater acceptability of pursuing a planned, though potentially higher-risk, pregnancy. We now have clear frameworks for pregnancy planning and management, but preterm birth and operative deliveries remain significantly increased. Preeclampsia continues to cause major maternal and neonatal morbidity and mortality worldwide, as delivery remains the only effective treatment. Pregnancy-related AKI remains a major clinical issue in many parts of the world. Women's reproductive health should be prioritized in future research efforts to ensure equitable care.

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QUESTIONS

- Which one of the following statements is correct regarding the normal physiologic changes of pregnancy?
 - Glomerular filtration rate decreases.
 - Serum sodium increases to 4–5 mEq/L above non-pregnancy levels.
 - The plasma uric acid concentration increases.
 - There is a mild respiratory alkalosis, with a compensatory increase in renal excretion of bicarbonate.

Answer: d

Rationale: Pregnancy is characterized by an increase in minute ventilation and mild respiratory alkalosis, with a fall in the partial pressure of carbon dioxide to approximately 27–32 mm Hg. There is an appropriate renal response, with increased excretion of bicarbonate. The glomerular filtration rate increases by approximately 40% in pregnancy. This leads to a decrease in uric acid concentration in normal pregnancy. Serum sodium falls, leading to mild hyponatremia due to a decrease in the osmotic threshold for antidiuretic hormone release.

- Which one of the following best describes the clinical syndrome of preeclampsia?
 - Proteinuria is necessary to establish a diagnosis of preeclampsia.
 - Kidney biopsy in preeclampsia is characterized by immune complexes within the glomeruli.
 - Glomerular endotheliosis, a hallmark of preeclampsia, occurs due to impairment of vascular endothelial growth factor (VEGF) signaling.
 - Patients with a history of preeclampsia do not have increased incidence of cardiovascular diseases in the long term.

Answer: c

Rationale: The syndrome of preeclampsia is characterized by high circulating levels of soluble VEGF receptor (soluble fms-like tyrosine kinase-1 [sFLT1]) that induces glomerular capillary damage and endothelial swelling (glomerular endotheliosis) by inhibiting podocyte-derived VEGF signaling in the glomerulus. Proteinuria is not essential for preeclampsia diagnosis, although it is frequently a feature reflecting the glomerular pathology. Kidney biopsy is not indicated in preeclampsia unless there is a strong uncertainty regarding the etiology of glomerular pathology; immune complexes are not a feature. Preeclampsia is an important recognized risk factor for future cardiovascular disease, kidney disease, and hypertension.

- A 29-year-old woman with a history of hypertension and lupus nephritis presents to you for routine review. She received induction therapy for lupus nephritis approximately 1 year ago with mycophenolate mofetil and prednisolone and is now in remission. She informs you she is currently at 8 weeks' gestation, having had a fetal scan with her primary care provider. Her medications include mycophenolate 500 mg twice daily, hydroxychloroquine 200 mg daily, nifedipine (slow release) 60 mg daily, aspirin 81 mg daily, and prednisone 5 mg daily. Her blood pressure is 140/85 mm Hg. Physical examination

is unremarkable. Laboratory studies show serum creatinine 1.2 mg/dL (stable from prior measurements) and random urine protein-to-creatinine ratio 143 mg/g.

Which of the following is the MOST appropriate medication change?

- Discontinue nifedipine and start methyldopa.
- Decrease nifedipine to 30 mg daily.
- Discontinue hydroxychloroquine and start azathioprine.
- Discontinue mycophenolate and start azathioprine.
- Discontinue aspirin.

Answer: d

Rationale: Mycophenolate mofetil and mycophenolic acid, which are commonly used for the treatment of glomerular disease and kidney transplantation, are teratogenic and contraindicated in pregnancy. These agents should be discontinued at least 6 (preferably 12) weeks before conception when pregnancy is anticipated; women of childbearing age who receive these medications should be counseled to use highly effective birth control. Women who conceive pregnancy while taking these agents should be switched immediately to agents compatible with pregnancy and then counseled regarding the impact of first trimester exposure so that they can determine whether to continue with the pregnancy. For the maintenance of remission in lupus nephritis, azathioprine and prednisone are safe and effective in pregnancy. Discontinuation of hydroxychloroquine is not recommended because this agent reduces risk for lupus flare in pregnancy and fetal complications. Her blood pressure is at target on her current dose of nifedipine, and this agent is safe and effective in pregnancy, so no changes are indicated. Low-dose aspirin reduces the risk of preeclampsia and should be initiated (or continued) in women at increased risk for preeclampsia. The dose should be increased to 100 to 150 mg daily.

- A 36-year-old woman with chronic hypertension presents to your office after a positive pregnancy test. Ultrasound confirms 10-week singleton gestation. Before pregnancy she was taking perindopril with hydrochlorothiazide, which she stopped when she noted her positive pregnancy test. Her blood pressure in the office is 155/95 mm Hg. Which of the following would be the most appropriate next step?

- Start labetalol.
- Continue perindopril alone.
- Continue hydrochlorothiazide alone.
- Start atenolol.

Answer: a

Rationale: Labetalol is a safe and appropriate antihypertensive agent for use in pregnancy. Methyldopa and dihydropyridine calcium channel blockers, such as long-acting nifedipine, are also appropriate for use in pregnancy. Angiotensin-converting enzyme inhibitors and angiotensin receptor antagonists are contraindicated in pregnancy because they cause birth defects with second- and third-trimester exposure. Hydrochlorothiazide and other loop and thiazide diuretics are not teratogenic

but can theoretically interfere with the normal plasma volume expansion of pregnancy and are generally avoided. Atenolol has been associated with impaired fetal growth in some studies; agents with combined α - and β -blockade, such as labetalol, maintain placental perfusion and are not associated with this adverse effect.

5. A 28-year-old woman presents at 32 weeks' gestation with a new onset of edema in her hands and face. Her blood pressure, previously normal, is 168/105 mm Hg. Blood pressure repeated 6 hours later is 150/98 mm Hg. Physical examination is notable for edema in her upper and lower extremities bilaterally. Which of the following findings would establish the diagnosis of preeclampsia?
 - a. Uric acid >8 mg/dL
 - b. Urine protein excretion >150 mg on a 24-hour urine collection
 - c. Total bilirubin >2.0 mg/dL

- d. Platelet count <100/mm³
- e. Development of pedal edema

Answer: d

Rationale: Current criteria for the diagnosis of preeclampsia in women with the new onset of hypertension after 20 weeks' gestation include features of organ involvement (see [Table 58.2](#)) including proteinuria, thrombocytopenia, elevated liver enzymes, acute kidney injury, cerebral, or visual symptoms. Hyperuricemia is a common feature of preeclampsia but is not reliable for diagnosis and is not part of the diagnostic criteria. Transaminitis is a diagnostic feature of preeclampsia, but hyperbilirubinemia is not. Although pedal edema can be a symptom of preeclampsia, it is not considered a diagnostic criterion for preeclampsia.